

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
DEPARTMENT OF ELECTRONICS



BIGPROJECT REPORT

**Thiết kế mạch đo nhiệt độ sử dụng
Thermistor**

SUPERVISOR: Nguyễn Trung Hiếu

SUBJECT: Applied Electronics (EE3129)

GROUP: 02

List of Members

STT	MSSV	Họ Và Tên	Lớp
1	2212904	Đoàn Ngọc Sang	L02
2	2210780	Nguyễn Đại Đồng	L02
3	2210164	Trần Nguyễn Trâm Ánh	L02

Ho Chi Minh, .././20..

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Thiết kế mạch đo nhiệt độ sử dụng Thermistor. Mạch có tầm đo $20 - 80^{\circ}\text{C}$, độ phân giải 0.1°C , ngõ ra là điện áp tương ứng từ $0 - 5\text{V}$:

1. Lựa chọn và mô phỏng Thermistor 2 dây (có điện trở dây dẫn là 10Ω).
2. Lựa chọn điện trở song song để tuyến tính hóa.
3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet.
4. Đảm bảo sai số (giữa giá trị lý thuyết và thực tế đo qua mạch) $\pm 5\%$.
5. Thiết kế mạch đọc điện áp cho ngõ ra ADC. Tiến hành lựa chọn ADC (số bit, nguồn).
Mô phỏng mạch đọc và kiểm chứng.

Chapter 1. Lựa chọn và Mô phỏng Thermistor 2 dây

Với Thermistor 2 dây, với mỗi dây có điện trở dây dẫn là $10\ \Omega$, với tầm đo là $20^{\circ}\text{C} \rightarrow 80^{\circ}\text{C}$ và span là 60°C . Nhóm em chọn con Thermistor NTC là **NTCLE100E3** với các thông số tổng quan như sau:



LINKS TO ADDITIONAL RESOURCES


[3D Models](#)

[Design Tools](#)

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QUICK REFERENCE DATA		
PARAMETER	VALUE	UNIT
Resistance value at 25 °C	3.3 to 470K	Ω
Tolerance on R_{25} -value	± 2 ; ± 3 ; ± 5	%
$B_{25/85}$ -value	2880 to 4570	K
Tolerance on $B_{25/85}$ -value	± 0.5 to ± 3	%
Operating temperature range: At zero power dissipation; continuously At zero power dissipation; for short periods	-40 to +125 ≤ 150	$^{\circ}\text{C}$
Response time (in oil)	≈ 1.2	s
Thermal time constant τ (for information only)	15	s
Dissipation factor δ (for information only)	7 8.5 (for R_{25} -value $\leq 680\ \Omega$)	mW/K
Maximum power dissipation at 55 °C	500	mW
Climatic category (LCT / UCT / days)	40 / 125 / 56	-
Weight	≈ 0.3	g

Hình 1.1: Bảng Datasheet tổng quát của NTCLE100E3.

Chapter 2. Lựa chọn điện trở song song để tuyến tính hóa

Theo datasheet của NTCLE100E3,

$$R_{(T)} = R_{ref} \times e^{A+B/T^1+C/T^2+D/T^3} \quad (2.1)$$

$$T_{(R)} = \left(A_1 + B_1 \ln^2 \frac{R}{R_{ref}} - C_1 \ln^2 \frac{R}{R_{ref}} - D_1 \ln^3 \frac{R}{R_{ref}} \right)^{-1}$$

Trong đó:

- $A, B, C, D, A_1, B_1, C_1$ và D_1 là các hằng số dựa vào thông số của vật liệu.
- R_{ref} là giá trị điện trở tại nhiệt độ tham chiếu (thường chọn ở $T = 25^\circ C$, $R_{ref} = R_{25}$).
- T là nhiệt độ Kelvin (K). $T(^{\circ}C) = T(K) - 273.15$.

Từ phương trình 2.1,

$$R_{TH} = R_{25} \times e^{A+B/T^1+C/T^2+D/T^3}$$

$$V_e = I \times \frac{R_{TH} \times R_p}{R_{TH} + R_p}$$

Ta có:

$$\begin{aligned} \frac{dV_e}{dT} &= \frac{I \times R_p \left(R_{TH} \left(-\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right) (R_{TH} + R_p) - R_{TH} \left(-\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right) R_{TH} \right)}{(R_{TH} + R_p)^2} \\ &= \frac{I R_p^2 R_{TH} \left(-\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4} \right)}{(R_{TH} + R_p)^2} \\ \frac{d^2 V_e}{dT^2} &= I R_p^2 \left(\frac{R_{TH} (A_1^2 + A_2) (R_{TH} + R_p)^2 - 2(R_{TH} + R_p) R_{TH}^2 A_1^2}{(R_{TH} + R_p)^4} \right) \end{aligned}$$

Trong đó,

$$- A_1 = -\frac{B}{T^2} - \frac{2C}{T^3} - \frac{3D}{T^4}$$

$$- A_2 = \frac{2B}{T^3} + \frac{6C}{T^4} + \frac{12D}{T^5}$$

$$\Rightarrow \frac{d^2 V_e}{d^2 T} = I R_p^2 \left(\frac{R_{TH}(A_1^2 + A_2)(R_{TH} + R_p) - 2R_{TH}^2 A_1^2}{(R_{TH} + R_p)^3} \right)$$

Từ Datasheet của NTCLE100E3 với $R_{25} = 10 \text{ k}\Omega$:

ELECTRICAL DATA AND ORDERING INFORMATION					SAP MATERIAL AND ORDERING NUMBER ⁽¹⁾	
R_{25} (Ω)	R_{25} -TOL. ($\pm$ %)	$B_{25/85}$ (K)	$B_{25/85}$ -TOL. ($\pm$ %)	UL RECOG. 	NTCLE100E3...B0/T1/T2 RoHS-COMPLIANT WITH EXEMPTION ⁽²⁾	NTCLE100E3...B0A/T1A/T2A RoHS-COMPLIANT
330	5	3560	1.5	✓	331*B0	331*B0A
470	5	3560	1.5	✓	471*B0	471*B0A
680	5	3560	1.5	✓	681*B0	681*B0A
1000	2, 3, 5	3528	0.5	✓	102*B0	102*B0A
1500	2, 3, 5	3528	0.5	✓	152*B0	152*B0A
2000	2, 3, 5	3528	0.5	✓	202*B0	202*B0A
2200	2, 3, 5	3977	0.75	✓	222*B0	222*B0A
2700	2, 3, 5	3977	0.75	✓	272*B0	272*B0A
3300	2, 3, 5	3977	0.75	✓	332*B0	332*B0A
4700	2, 3, 5	3977	0.75	✓	472*B0	472*B0A
5000	2, 3, 5	3977	0.75	✓	502*B0	502*B0A
6800	2, 3, 5	3977	0.75	✓	682*B0	682*B0A
10 000	2, 3, 5	3977	0.75	✓	103*B0	103*B0A
12 000	2, 3, 5	3740	2	✓	123*B0	123*B0A
15 000	2, 3, 5	3740	2	✓	153*B0	153*B0A
22 000	2, 3, 5	3740	2	✓	223*B0	223*B0A
33 000	2, 3, 5	4090	1.5	✓	333*B0	333*B0A
47 000	2, 3, 5	4090	1.5	✓	473*B0	473*B0A
50 000	2, 3, 5	4190	1.5	✓	503*B0	503*B0A
68 000	2, 3, 5	4190	1.5	✓	683*B0	683*B0A
100 000	2, 3, 5	4190	1.5	✓	104*B0	104*B0A
150 000	2, 3, 5	4370	2.5	✓	154*B0	154*B0A
220 000	2, 3, 5	4370	2.5	✓	224*B0	224*B0A
330 000	2, 3, 5	4570	1.5		334*B0	334*B0A
470 000	2, 3, 5	4570	1.5		474*B0	474*B0A

Ta chọn được $B_{25/85} = 3977$. Từ đó ta chọn được các giá trị của A , B , C và D như sau:

$$- A = -14.6337$$

$$- B = 4791,842$$

$$- C = -115334$$

$$- D = -3730535$$

PARAMETER FOR DETERMINING NOMINAL RESISTANCE VALUES											
NUMBER	B _{25/85} (K)	NAME	TOL. B (%)	A	B (K)	C (K ²)	D (K ³)	A ₁	B ₁ (K ⁻¹)	C ₁ (K ⁻²)	D ₁ (K ⁻³)
1	2880	Mat O. with Bn = 2880K	3	- 9.094	2251.74	229098	- 2.744820E+07	3.354016E-03	3.495020E-04	2.095959E-06	4.260615E-07
2	2990	Mat P. with Bn = 3990K	3	- 10.2296	2887.62	132336	- 2.502510E+07	3.354016E-03	3.415560E-04	4.955455E-06	4.364236E-07
3	3041	Mat Q. with Bn = 3041K	3	- 11.1334	3658.73	- 102895	5.166520E+05	3.354016E-03	3.349290E-04	3.683843E-06	7.050455E-07
4	3136	Mat R. with Bn = 3136K	3	- 12.4493	4702.74	- 402687	3.196830E+07	3.354016E-03	3.243880E-04	2.658012E-06	- 2.701560E-07
5	3390	Mat S. with Bn = 3390K	3	- 12.6814	4391.97	- 232807	1.509643E+07	3.354016E-03	2.993410E-04	2.135133E-06	- 5.672000E-09
6	3528 ⁽¹⁾	Mat I. with Bn = 3528K	0.5	- 12.0596	3687.667	- 7617.13	- 5.914730E+06	3.354016E-03	2.909670E-04	1.632136E-06	7.192200E-08
	3528 ⁽²⁾			- 21.0704	11903.95	- 2504699	2.470338E+08	3.354016E-03	2.933908E-04	3.494314E-06	- 7.712690E-07
7	3560	Mat H. with Bn = 3560K	1.5	- 13.0723	4190.574	- 47158.4	- 1.199256E+07	3.354016E-03	2.884193E-04	4.118032E-06	1.786790E-07
8	3740	Mat B. with Bn = 3740K	2	- 13.8973	4557.725	- 98275	- 7.522357E+06	3.354016E-03	2.744032E-04	3.666944E-06	1.375492E-07
9	3977	Mat A. with Bn = 3977K	0.75	- 14.6337	4791.842	- 115334	- 3.730535E+06	3.354016E-03	2.569850E-04	2.620131E-06	6.383091E-08
10	4090	Mat C. with Bn = 4090K	1.5	- 15.5322	5229.973	- 160451	- 5.414091E+06	3.354016E-03	2.519107E-04	3.510939E-06	1.105179E-07
11	4190	Mat D. with Bn = 4190K	1.5	- 16.0349	5459.339	- 191141	- 3.328322E+06	3.354016E-03	2.460382E-04	3.405377E-06	1.034240E-07
12	4370	Mat E. with Bn = 4370K	2.5	- 16.8717	5759.15	- 194267	- 6.869149E+06	3.354016E-03	2.367720E-04	3.585140E-06	1.255349E-07
13	4570	Mat F. with Bn = 4570K	1.5	- 17.6439	6022.726	- 203157	- 7.183526E+06	3.354016E-03	2.264097E-04	3.278184E-06	1.097628E-07

Notes

⁽¹⁾ Temperature < 25 °C

⁽²⁾ Temperature ≥ 25 °C

$$\Rightarrow \begin{cases} A_1 = -0.03802560045 \\ A_2 = 2.078383337 \end{cases} \quad \text{Tại } T = 50^\circ\text{C} = 323.15\text{K}$$

$$\text{Xét } \frac{V^2 V_e}{d^2 T} = 0 \quad \text{Tại } T = 50^\circ\text{C} = 323.15\text{K}.$$

Từ Datasheets,

RESISTANCE VALUES AT INTERMEDIATE TEMPERATURES WITH R_{25} AT (2.2, 2.7, 3.3, 4.7, 5.0, 6.8, 10) k Ω								
T _{OPER} (°C)	PART NUMBER NTCLE100E322*** R_T (Ω)	PART NUMBER NTCLE100E3272*** R_T (Ω)	PART NUMBER NTCLE100E3332*** R_T (Ω)	PART NUMBER NTCLE100E3472*** R_T (Ω)	PART NUMBER NTCLE100E3502*** R_T (Ω)	PART NUMBER NTCLE100E3682*** R_T (Ω)	PART NUMBER NTCLE100E3103*** R_T (Ω)	TCR (%/K) $\Delta R/R$ DUE TO B_{tot} (%)
-40	73 061	89 665	109 591	156 084	166 047	225 824	332 094	-6.62 2.79
-35	52 778	64 773	79 167	112 753	119 950	163 132	239 900	-6.39 2.52
-30	38 544	47 304	57 816	82 344	87 600	119 136	175 200	-6.18 2.26
-25	28 443	34 907	42 665	60 765	64 643	87 915	129 287	-5.98 2.02
-20	21 199	26 017	31 798	45 288	48 179	65 524	96 358	-5.78 1.78
-15	15 950	19 575	23 925	34 075	36 250	49 300	72 500	-5.60 1.55
-10	12 110	14 862	18 165	25 872	27 523	37 431	55 046	-5.42 1.33
-5	9275	11 382	13 912	19 814	21 078	28 667	42 157	-5.25 1.12
0	7162	8790	10 743	15 300	16 277	22 137	32 554	-5.09 0.92
5	5574	6841	8362	11 909	12 669	17 230	25 339	-4.93 0.72
10	4372	5365	6558	9340	9936	13 513	19 872	-4.79 0.53
15	3454	4239	5180	7378	7849	10 675	15 698	-4.64 0.35
20	2747	3372	4121	5869	6244	8492	12 488	-4.51 0.17
25	2200	2700	3300	4700	5000	6800	10 000	-4.38 0.00
30	1773	2176	2659	3788	4030	5480	8059	-4.25 0.17
35	1438	1764	2156	3071	3267	4444	6535	-4.13 0.32
40	1173	1439	1759	2505	2665	3624	5330	-4.02 0.48
45	961.8	1180	1443	2055	2186	2973	4372	-3.94 0.63
50	793.2	973.4	1190	1694	1803	2452	3605	-3.80 0.77
55	657.5	806.9	986.3	1405	1494	2032	2989	-3.70 0.91
60	547.8	672.3	821.7	1170	1245	1693	2490	-3.60 1.05

Ta có $R_{50} = 3605 \Omega$.

$$\Rightarrow R_p^2 R_{TH} (A_1^2 + A_2) (R_{TH} + R_p) - 2 R_p^2 R_{TH}^2 A_1^2 = 0$$

$$\Leftrightarrow R_p^2 R_{50} (A_1^2 + A_2) (R_{50} + R_p) - 2 R_p^2 R_{50}^2 A_1^2 = 0$$

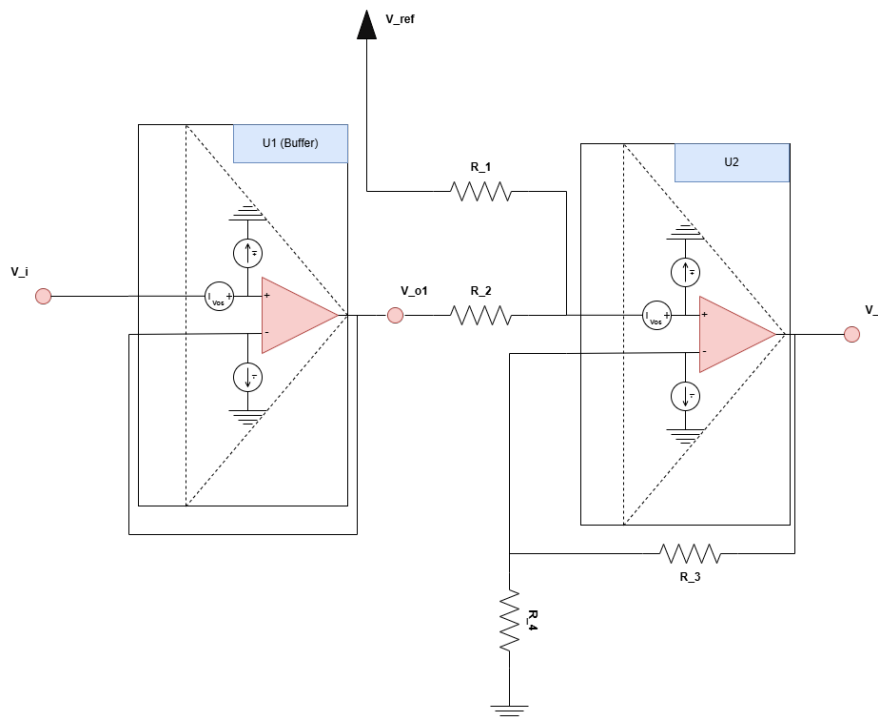
$$\Leftrightarrow R_p^3 5.962 - 16090 R_p^2 = 0$$

$$\Rightarrow R_p = 2.698 \text{ k}\Omega \approx 2.7 \text{ k}\Omega$$

Chapter 3. Lựa chọn OPAMP sử dụng, mô phỏng OPAMP theo datasheet

3.1 Tại tầng khuếch đại

Ta xét ở tầng khuếch đại:



1. Xét tại OPAMP đầu tiên có chức năng là một Buffer:

$$V_{o1} = V_i \pm V_{os}$$

2. Xét tại OPAMP thứ hai có chức năng là thực hiện hàm $V_o = V_i + V_{ref}$:

$$V_o = \left(1 + \frac{R_3}{R_4}\right) \left(V_{ref} \frac{R_2}{R_1 + R_2} + V_{o1} \frac{R_1}{R_1 + R_2}\right) \pm V_{os} \left(1 + \frac{R_3}{R_4}\right) \mp I_- R_3$$

Chọn $R_1 = R_2 = 1 \text{ k}\Omega$

Chọn $R_3 = R_4 = 1 \text{ k}\Omega$

Ta xét 2 con OPAMP trên là cùng một Package, ta có:

1. Xét tại OPAMP đầu tiên có chức năng là một Buffer với ngõ điện áp ngõ ra sai số 5% giá trị thì tại đây nhóm em chọn sai số phải rơi vào dưới 1%:

$$\pm V_{os} < 0.08 \text{ V}$$

$\Rightarrow V_{os}$ phải có giá trị ở dưới 10 lần ngưỡng trên là cỡ 8 mV.

2. Xét tại OPAMP thứ hai phải có ngõ ra cũng rơi vào dưới 1%:

$$\pm V_{os} \left(1 + \frac{R_3}{R_4} \right) \mp I_- R_3 < 0.05 \text{ V}$$

$\Rightarrow$ Với $V_{os} < 10 \text{ mV}$ thì $I_- > 70 \mu\text{A}$.

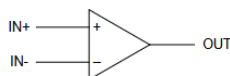
$$\Rightarrow \begin{cases} I_{os} = |I_+ - I_-| < 30 \mu\text{A} \\ I_{ib} = \frac{I_+ + I_-}{2} < 15 \mu\text{A} \end{cases}$$

Từ đó ở tầng khuếch đại nhóm em sử dụng **TL082CP**, với thông số sau:

Device Information		
PART NUMBER	CHANNEL COUNT	PACKAGE ⁽¹⁾
TL081x	Single	P (PDIP, 8)
		DCK (SC70, 5)
		PS (SO, 8)
		D (SOIC, 8)
		DBV (SOT-23, 5)
TL082x	Dual	P (PDIP, 8)
		PS (SO, 8)
		D (SOIC, 8)
		DDF (SOT-23, 8)
		PW (TSSOP, 8)
TL082M	Dual	JG (CDIP, 8)
		FK (LCCC, 20)
TL084x	Quad	N (PDIP, 14)
		D (SOIC, 14)
		DYY (SOT-23, 14)
		PW (TSSOP, 14)
TL084M	Quad	J (CDIP, 14)
		FK (LCCC, 20)

(1) For more information, see [Section 11](#).

TL081 (Each Amplifier)
TL082 (Each Amplifier)
TL084 (Each Amplifier)



5.7 Electrical Characteristics: TL08xH

For $V_S = (V_{CC+}) - (V_{CC-}) = 4.5\text{ V to }40\text{ V}$ ($\pm 2.25\text{ V to } \pm 20\text{ V}$) at $T_A = 25^\circ\text{C}$, $R_L = 10\text{ k}\Omega$ connected to $V_S / 2$, $V_{CM} = V_S / 2$, and $V_{out} = V_S / 2$, unless otherwise noted.

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
OFFSET VOLTAGE							
V _{OS}	Input offset voltage		T _A = −40°C to +125°C		±1	±4	mV
						±5	
dV _{OS} /dT	Input offset voltage drift		T _A = −40°C to +125°C		±2		µV/°C
PSRR	Input offset voltage versus power supply	V _S = 5V to 40V, V _{CM} = V _S /2	T _A = −40°C to +125°C		±1	±10	µV/V
	Channel separation	f = 0 Hz			10		µV/V
INPUT BIAS CURRENT							
I _B	Input bias current				±1	±120	pA
			DCK and DBV packages		±1	±300	pA
			T _A = −40°C to +125°C (1)			±5	nA
I _{OS}	Input offset current				±0.5	±120	pA
			DCK and DBV packages		±0.5	±250	pA
			T _A = −40°C to +125°C (1)			±5	nA

Chapter 4. Thực hiện mạch đọc điện áp cho ngõ ra ADC

4.1 Tổng quan về ADC

Theo chuẩn ngõ ra điện áp đọc từ Sensor là từ $0 \rightarrow 5\text{ V}$, với ngõ vào là từ $20^\circ\text{C} \rightarrow 80^\circ\text{C}$ với bước nhảy là 0.1°C . Thực hiện tính giá trị của một bước nhảy nhỏ nhất là:

$$\frac{5 - 0}{\frac{80 - 20}{0.1} + 1} \approx 8.3195\text{ mV}$$

Vậy có nghĩa là $\Delta_{min} \approx 8.3195\text{ mV}$, từ đó ta có:

$$2^Q = \frac{V_{ref+} - V_{ref-}}{\Delta_{min}}$$

Giả sử, chọn $V_{ref+} = 5\text{ V}$ và $V_{ref-} = 0\text{ V}$. Ta có $Q \approx 9.23\text{ bit}$

Tuy nhiên, trong thị trường không có ADC 9.23 bit cho nên nhóm em sẽ chọn con có trên thị trường là **ADS1115IDGSR**.

4.2 Electrical Characteristics của ADS1115

- Ultra-small packages:
 - + X2QFN: $2\text{ mm} \times 1.5\text{ mm} \times 0.4\text{ mm}$
 - + SOT: $2.9\text{ mm} \times 2.8\text{ mm} \times 0.6\text{ mm}$
- Wide supply range: 2.0 V to 5.5 V
- Low current consumption: $150\text{ }\mu\text{A}$ (continuous-conversion mode)
- Programmable data rate: 8SPS to 860SPS
- Single-cycle settling
- Internal low-drift voltage reference
- Internal oscillator
- I2C interface: four pin-selectable addresses

- Operating temperature range: -40°C to $+125^{\circ}\text{C}$
- Family of devices:
 - + ADS1115: four single-ended or two differential inputs with comparator and PGA

Từ những thông tin, ADC ADS1115 có tầm hoạt động ở dải nhiệt độ phù hợp trong khoảng $20^{\circ}\text{C} \rightarrow 80^{\circ}\text{C}$, nhóm em sử dụng nguồn cấp là $V_{DD} = 5.5\text{V}$ và cho phép nguồn sai số trong khoảng từ $[5.3, 5.5]\text{V}$ phù hợp với thông số dưới đây:

DIGITAL INPUT/OUTPUT					
V_{IH}	High-level input voltage		0.7 VDD	5.5	V
V_{IL}	Low-level input voltage		GND	0.3 VDD	V
V_{OL}	Low-level output voltage	$I_{OL} = 3\text{mA}$	GND	0.15	0.4 V
	Input leakage current	$GND < V_{DIG} < VDD$	-10	10	μA

Hình 4.1: Digital Input/Output của ADS1115.

Như vậy, với $GND \approx 0\text{V} < V_{IN} < V_{DD} = 5.5\text{V}$ thì có thể phù hợp cho tầm đo từ $0 \rightarrow 5\text{V}$ từ mạch đọc giá trị của sensor.

Tiếp theo, ta cần quan tâm đến giá trị như sau:

SYSTEM PERFORMANCE				
	Resolution (no missing codes)		16	Bits
DR	Data rate		8, 16, 32, 64, 128, 250, 475, 860	SPS
	Data rate variation	All data rates	-10% 10%	
	Output noise		See Noise Performance section	
INL	Integral nonlinearity	DR = 8SPS, FSR = $\pm 2.048V^{(2)}$	1	LSB
	Offset error	FSR = $\pm 2.048V$, differential inputs	± 1	LSB
		FSR = $\pm 2.048V$, single-ended inputs	± 3	LSB
	Offset drift over temperature	FSR = $\pm 2.048V$	0.005	LSB/ $^{\circ}C$
	Long-term Offset drift	FSR = $\pm 2.048V$, $T_A = 125^{\circ}C$, 1000 hours	± 1	LSB
	Offset power-supply rejection	FSR = $\pm 2.048V$, DC supply variation	1	LSB/V
	Offset channel match	Match between any two inputs	3	LSB
	Gain error ⁽³⁾	FSR = $\pm 2.048V$, $T_A = 25^{\circ}C$	0.01% 0.15%	
	Gain drift over temperature ⁽³⁾	FSR = $\pm 0.256V$	7	ppm/ $^{\circ}C$
		FSR = $\pm 2.048V$	5 40	
		FSR = $\pm 6.144V^{(1)}$	5	
	Long-term gain drift ⁽³⁾	FSR = $\pm 2.048V$, $T_A = 125^{\circ}C$, 1000 hours	$\pm 0.05\%$	
	Gain power-supply rejection		80	ppm/V
	Gain match ⁽³⁾	Match between any two gains	0.02% 0.1%	
	Gain channel match	Match between any two inputs	0.05% 0.1%	
CMRR	Common-mode rejection ratio	At DC, FSR = $\pm 0.256V$	105	dB
		At DC, FSR = $\pm 2.048V$	100	
		At DC, FSR = $\pm 6.144V^{(1)}$	90	
		$f_{CM} = 60Hz$, DR = 8SPS	105	
		$f_{CM} = 50Hz$, DR = 8SPS	105	

Hình 4.2: Electrical Characteristics của ADS1115.

- Data rate (DR) là giá trị dữ liệu Digital được đổi theo đơn vị thời gian. Với DR = 128 Sample per second là giá trị default của ADS1115, với thời gian cách nhau đọc dữ liệu khoảng tầm 8 ms.
- Integral Nonlinearity (INL) là thước đo độ lệch của đặc tuyến chuyển đổi thực tế so với đặc tuyến lý tưởng trong các bộ chuyển đổi ADC/DAC, sau khi đã loại bỏ sai số offset và gain. Với INL = 1 LSB.
- Offset error là sai số khi sử dụng chế độ Single-Ended với giá trị là ± 3 LSB.
- Gain drift over temperature là sai số của phép đo khi có ảnh hưởng của nhiệt độ với tại FSR = $\pm 6.144V$ thì Gain drift = $5ppm/^{\circ}C$ tức là mỗi $^{\circ}C$ sẽ gây sai số là 5 phần ngàn.
- Common-mode Rejection Ratio (CMRR) là thước đo khả năng của một bộ khuếch đại vi sai trong việc loại bỏ tín hiệu chung xuất hiện đồng thời và cùng pha trên cả hai đầu vào. Với CMRR = 90 dB.

4.3 Typical Characteristics của ADS1115

- Single-Ended Offset Error vs Temperature:

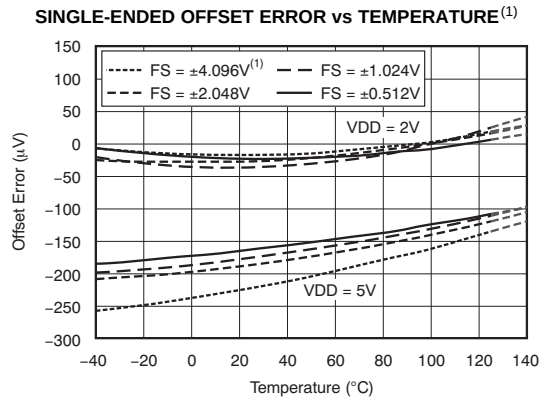


Figure 4.

+ Tại $T = 20^{\circ}\text{C}$: Offset Error = $-225\mu\text{V}$

+ Tại $T = 80^{\circ}\text{C}$: Offset Error = $-120\mu\text{V}$

- Gain Error vs Temperature:

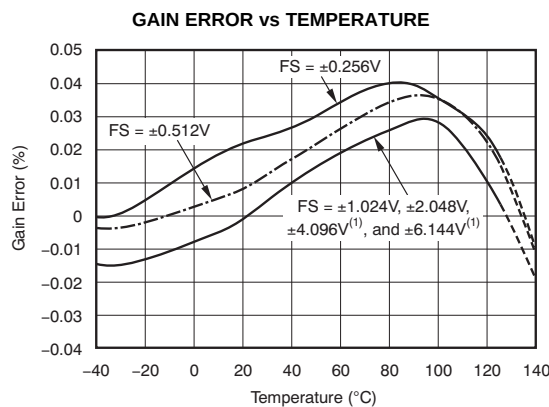


Figure 6.

+ Tại $T = 20^{\circ}\text{C}$: Gain Error = 0.00 %

+ Tại $T = 80^{\circ}\text{C}$: Gain Error = 0.03 %

- Gain Error vs Supply:

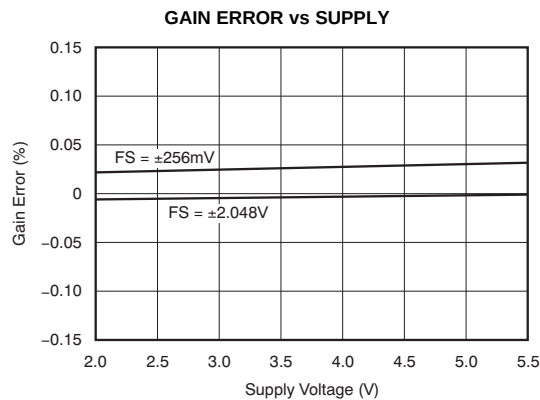


Figure 7.

+ Tại $V_{DD} = 5.5\text{V}$: Gain Error = 0.00 %

- INL vs Supply Voltage:

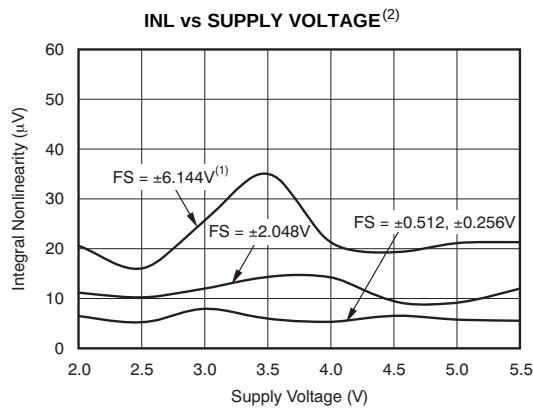


Figure 8.

+ Tại $V_{DD} = 5.5V$: INL vs Voltage = $20\mu V$

- INL vs Temperature:

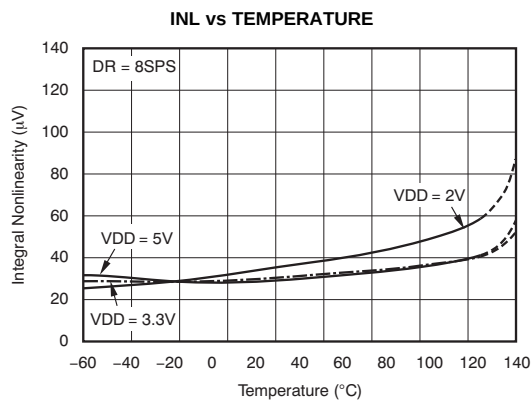


Figure 13.

+ Tại $T = 20^{\circ}C$: INL vs Temperature = $22\mu V$

+ Tại $T = 80^{\circ}C$: INL vs Temperature = $38\mu V$

- Data Rate vs Temperature:

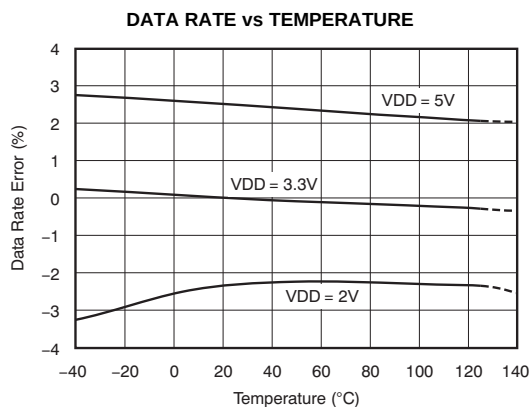


Figure 20.

+ Tại $T = 20^{\circ}C$: Data Rate Error = 2.8%

+ Tại $T = 80^{\circ}C$: Data Rate Error = 1.2%

4.4 Data Format

Với ADS1115 là một ADC 16-bit với ngõ ra là một chuỗi bit có dấu theo giá trị như sau:

Table 4. Input Signal versus Ideal Output Code

INPUT SIGNAL, V_{IN} ($A_{INP} - A_{INN}$)	IDEAL OUTPUT CODE ⁽¹⁾
$\geq FS (2^{15} - 1)/2^{15}$	7FFFh
$+FS/2^{15}$	0001h
0	0
$-FS/2^{15}$	FFFFh
$\leq -FS$	8000h

1. Excludes the effects of noise, INL, offset, and gain errors.

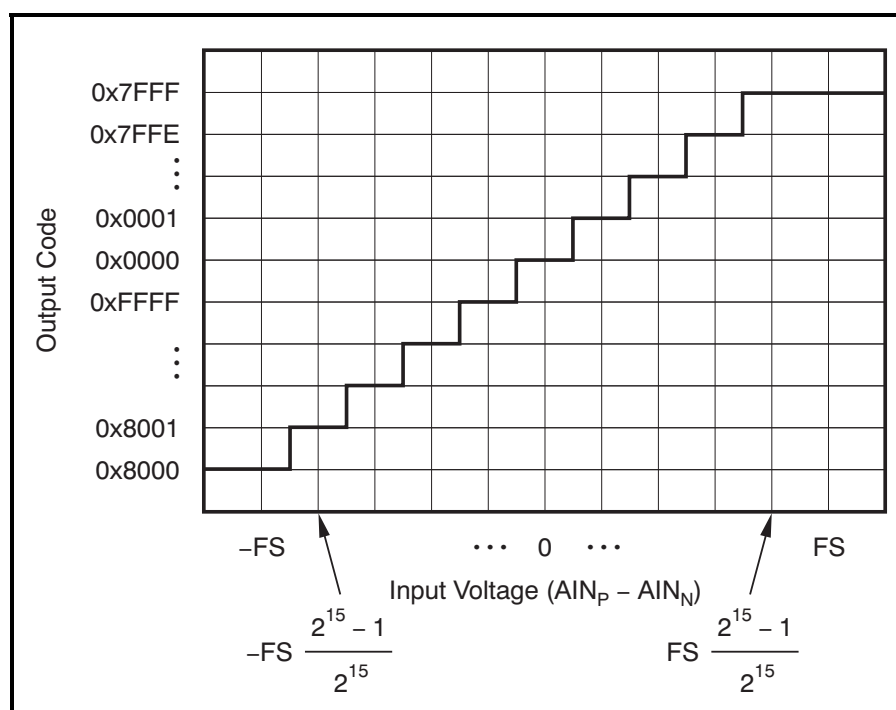


Figure 27. ADS1113/4/5 Code Transition Diagram

Hình 4.3: ADS1115 Code Transition Diagram.

Theo hình 4.3, với $FS = 6.144$ như ở trên thì giá trị lớn nhất mà ADC có thể đọc được

là $V_{ref} \times \frac{2^{15} - 1}{2^{15}} \approx 6.1438 \text{ V}$ vẫn phù hợp với tầm đo là từ $0 \rightarrow 5 \text{ V}$. Từ đó ta có công thức suy ngược từ D_{out} nhận được từ ADC là:

$$V_{out} = \frac{V_{ref}}{2^{15}} \times D_{out} = \frac{6.1438}{32768} \times D_{out}$$

4.5 Kết quả của việc tính toán ADC

Ta có bảng tính toán giá bên dưới, trong bảng giá trị có các giá trị như sau:

- **T_s** : là giá trị nhiệt độ chuẩn cần đo.
- **V_i** : là giá trị điện áp đo được từ sensor trong khoảng từ $0 \rightarrow 5 \text{ V}$ với sai số là 5%.
- **Offset_Error** : là giá trị sai số của ADS1115 do nhiệt độ tạo nên.
- **Gain_Error** : là giá trị của độ lợi của ADS1115 do nhiệt độ tạo nên.
- **V_i_ADC** : là giá trị mà ADC thực sự nhận tại chân IN0.
- **D_out** : là giá trị tính toán từ giá trị điện đọc từ chân IN0 với công thức là $\frac{V_{i_ADC} \times 2^{15}}{FS}$.
- **D_cal_Error** : là giá trị mà thực sự ADS1115 cho ra ở ngõ ra với các giá trị sai số cộng thêm do chính ADC và nhiệt độ.
- **Cal_V** : là giá trị được tính trong MCU chuyển đổi từ giá trị Digital của ADS1115 gửi về cho MCU.
- **Cal_Temp** : là giá trị chuyển đổi từ điện áp tính toán được **Cal_V** về giá trị nhiệt độ theo công thức chuẩn là $T = \frac{V_{in} + \frac{5}{3}}{\frac{1}{12}}$.

Requirement			
ERROR	5%		

ADC				
	Min	Typ	Max	Unit
VDD	5.2		5.5	5.5 V
GND	-0.3		0	0.3 V
Resolution			16	Bit
Analog_In_V_to_GND				V
Temperature	-60		150	dó C
PGA (program gain amplifier)	0.66666667	0.66686667		V
FS (Full-Scale)	-6.142157353	6.1438125	6.142157353	V
Common-Mode Input Impedance		6000000		Ohm
DR (Data Rate)		128		SPS
Range_Temperature	20		80	dó C
Offset Voltage	-0.000225		-0.00012	V
Gain Error	0%		0.03%	%
Gain Drift		0.000500%		%/dó C
	Tai FS = 6.144	0.030000%		
Common Mode Rejection	Tai FS = 6.144	90		dB
INL_vs_VoltageSupply		0.000020		V
INL_vs_Temperature	0.000022		0.000038	V
Digital Input/Output				
V_IH	3.85		5.5	V
V_IL	-0.5		1.65	V

Calibration			
	Value_Standard	Value_Measurement	Error
raw_zero_25	0	0	0
raw_full_25	5	5	0
raw_zero_20	0	0	0
raw_full_20	5	5	0
raw_zero_50	0	0	0
raw_full_50	5	5	0
raw_zero_80	0	0	0
raw_full_80	5	5	0

Digital output			
Input Signal (V_i)	>= FS(2 ¹⁵ -1)/(2 ¹⁵)		
Offset Error	-3	6.141969909 V	3 LSB
Offset Drift		0.005	LSB/dó C
	FS = 6.144V	0.3	LSB
Offset power supply rejection		1	LSB/V
	VDD = 5.5V	0.3	LSB
INL		1	LSB

s	V	Offset_Vol	Gain_Error	G_E	V_I_ADC	Error_V_I_ADC	D_out	D_cat_error	BIN	Cal_V	Cal_Temp	Error_re	Error_ab
20	0	0.017	-0.000225	0.000000%	-0.001500	0.017257201	1.5129%	92.0688	94	0000000001011110	0.017624462	20.2114935	1%
20.1	0.019954682	-0.000224	0.000100%	-0.001469	0.020207083	1.2649%	107.8067	111	0000000001011111	0.020811865	20.2497424	1%	
20.2	0.025642402	-0.000224	0.000100%	-0.001439	0.025887233	0.9548%	138.1109	137	0000000010001001	0.025686716	20.3082406	1%	
20.3	0.032330742	-0.000223	0.000200%	-0.001408	0.032656971	0.7277%	173.7426	175	0000000010101111	0.032811499	20.393738	0%	
20.4	0.038074328	-0.000223	0.000200%	-0.001378	0.038303627	0.5996%	204.3482	202	0000000011001010	0.037873844	20.4544861	0%	
20.5	0.044816929	-0.000222	0.000300%	-0.001347	0.045036343	0.4896%	240.2732	240	0000000011110000	0.044998627	20.5399835	0%	
20.6	0.05158625	-0.000222	0.000300%	-0.001317	0.051798108	0.4107%	276.3479	280	0000000010001100	0.052498398	20.6299808	0%	
20.7	0.058382283	-0.000221	0.000400%	-0.001286	0.058586016	0.3490%	312.562	313	0000000010011001	0.058685709	20.7042285	0%	
20.8	0.065205052	-0.000220	0.000400%	-0.001256	0.065400949	0.3005%	348.9204	347	0000000010101011	0.065060514	20.7807262	0%	
20.9	0.072054452	-0.000220	0.000500%	-0.001225	0.072244024	0.2631%	385.4288	386	0000000011000010	0.072372791	20.8684735	0%	
21	0.078930568	-0.000219	0.000500%	-0.001195	0.079113097	0.2313%	422.076	423	0000000011010011	0.07931008	20.951721	0%	
21.1	0.084867512	-0.000219	0.000600%	-0.001164	0.085045587	0.2098%	453.7264	454	0000000011000110	0.085122402	21.0214688	0%	
21.2	0.091797781	-0.000218	0.000600%	-0.001134	0.091969553	0.1871%	490.6664	489	0000000011110100	0.091684702	21.1002164	0%	
21.3	0.098754705	-0.000218	0.000700%	-0.001103	0.098921658	0.1691%	527.7566	532	0000000010000101	0.099746956	21.1969635	0%	
21.4	0.105738272	-0.000217	0.000700%	-0.001073	0.105899704	0.1527%	564.9851	570	0000000010001110	0.106871738	21.2824609	1%	
21.5	0.112748471	-0.000216	0.000800%	-0.001042	0.112904885	0.1387%	602.3584	605	0000000010010110	0.113434038	21.3612085	1%	
21.6	0.118823527	-0.000216	0.000800%	-0.001012	0.118977181	0.1293%	634.7547	639	0000000010011111	0.119808844	21.4377061	1%	
21.7	0.125887774	-0.000215	0.000900%	-0.000981	0.126037188	0.1187%	672.4205	672	0000000010101000	0.125996155	21.5119539	1%	
21.8	0.132978615	-0.000215	0.000900%	-0.000951	0.133125068	0.1101%	710.235	709	0000000010110001	0.132933443	21.5952013	1%	
21.9	0.140096038	-0.000214	0.001000%	-0.000920	0.140239071	0.1021%	748.1889	749	0000000010110101	0.140433214	21.6851986	1%	
22	0.146281565	-0.000214	0.001000%	-0.000890	0.14642333	0.0969%	781.1824	785	0000000011000100	0.147183008	21.7661961	1%	
22.1	0.153452935	-0.000213	0.001100%	-0.000859	0.153592049	0.0907%	819.4283	819	0000000011001100	0.153557814	21.8426938	1%	
22.2	0.160650843	-0.000212	0.001100%	-0.000829	0.160787598	0.0851%	857.8173	857	0000000010101010	0.160682596	21.9281912	1%	
22.3	0.166919306	-0.000212	0.001200%	-0.000798	0.167056042	0.0819%	891.26	890	0000000011011101	0.166869907	22.0024389	1%	
22.4	0.174717107	-0.000211	0.001200%	-0.000768	0.174306255	0.0776%	929.9406	934	0000000011010010	0.175119656	22.1014359	1%	
22.5	0.181449339	-0.000211	0.001300%	-0.000737	0.181584554	0.0745%	968.7711	970	0000000011100101	0.18186945	22.1824334	1%	
22.6	0.187800619	-0.000210	0.001300%	-0.000707	0.187935789	0.0720%	1002.6555	1004	0000000011110100	0.188244255	22.2589311	2%	
22.7	0.195132651	-0.000209	0.001400%	-0.000676	0.195267689	0.0692%	1041.7719	1040	0000000010000000	0.194994049	22.3399286	2%	
22.8	0.202491122	-0.000209	0.001400%	-0.000646	0.202627262	0.0672%	1081.0359	1083	0000000010000011	0.203056303	22.4366756	2%	
22.9	0.208925084	-0.000208	0.001500%	-0.000615	0.209062546	0.0658%	1115.3688	1115	0000000010001011	0.20905612	22.5086734	2%	
23	0.216337223	-0.000208	0.001500%	-0.000585	0.216476618	0.0644%	1154.9236	1154	0000000010000010	0.216368397	22.5964208	2%	
23.1	0.222826509	-0.000207	0.001600%	-0.000554	0.222968017	0.0635%	1189.5558	1189	0000000010010010	0.222930696	22.6751684	2%	
23.2	0.230292244	-0.000207	0.001600%	-0.000524	0.230436526	0.0627%	1229.401	1228	0000000010011001	0.230242973	22.7629157	2%	
23.3	0.236836788	-0.000206	0.001700%	-0.000493	0.236983983	0.0622%	1264.3323	1265	0000000010011001	0.237180262	22.8461631	2%	
23.4	0.244356043	-0.000205	0.001700%	-0.000463	0.244505864	0.0613%	1304.4623	1308	0000000010000110	0.245242516	22.9429102	2%	
23.5	0.251901602	-0.000205	0.001800%	-0.000432	0.252055739	0.0612%	1344.7416	1347	0000000010100001	0.252554792	23.0306575	2%	
23.6	0.258528469	-0.000204	0.001800%	-0.000402	0.2586685	0.0611%	1380.1173	1379	0000000010101001	0.258554609	23.1026553	2%	
23.7	0.266127425	-0.000204	0.001900%	-0.000371	0.266290652	0.0613%	1420.6862	1424	0000000010010000	0.266991852	23.2039022	2%	
23.8	0.272809362	-0.000203	0.001900%	-0.000341	0.272977297	0.0616%	1456.3601	1460	0000000010110100	0.273741646	23.2848997	2%	
23.9	0.28046163	-0.000203	0.002000%	-0.000310	0.280635651	0.0620%	1497.2182	1497	0000000010110101	0.280678934	23.3681472	2%	
24	0.287198558	-0.000202	0.002000%	-0.000280	0.287378108	0.0625%	1533.1898	1533	0000000010111110	0.287428728	23.4491447	2%	
24.1	0.293963383	-0.000201	0.002100%	-0.000249	0.294149154	0.0632%	1569.314	1574	0000000010000101	0.295115993	23.5413919	2%	
24.2	0.300756084	-0.000201	0.002100%	-0.000219	0.300949189	0.0642%	1605.5929	1609	0000000010010010	0.301678293	23.6201395	2%	
24.3	0.307576642	-0.000200	0.002200%	-0.000188	0.307776791	0.0651%	1642.0188	1644	0000000010011010	0.308240593	23.6988871	2%	
24.4	0.315363091	-0.000200	0.002200%	-0.000158	0.315571724	0.0660%	1683.6029	1688	0000000010010000	0.316490341	23.7978841	2%	
24.5	0.32238425	-0.000199	0.002300%	-0.000127	0.32245448	0.0670%	1720.3257	1721	0000000010010101	0.322677652	23.8721318	3%	
24.6	0.329141548	-0.000199	0.002300%	-0.000097	0.329366605	0.0684%	1757.2025	1760	0000000010110000	0.329989929	23.9598792	3%	
24.7	0.337007873	-0.000198	0.002400%	-0.000066	0.337214619	0.0694%	1799.2165	1802	0000000010000101	0.337864689	24.0543763	3%	
24.8	0.3439563	-0.000197	0.002400%	-0.000036	0.344208241	0.0705%	1836.3841	1838	0000000010010110	0.344614483	24.1353738	3%	
24.9	0.350951106	-0.000197	0.002500%	-0.000005	0.351204351	0.0722%	1873.709	1877	0000000010101010	0.35192676	24.2231211	3%	
25	0.358897075	-0.000196	0.002500%	0.000025	0.359160052	0.0733%	1916.1534	1921	0000000011000001	0.360176508	24.3221181	3%	
25.1	0.365937033	-0.000196	0.002600%	0.000056	0.366211538	0.0750%	1953.7738	1954	0000000011010000	0.366363819	24.3963658	3%	
25.2	0.373004635	-0.000195	0.002600%	0.000086	0.373289733	0.0764%	1991.5366	1993	0000000011010010	0.373676096	24.4841132	3%	
25.3	0.381030001	-0.000195	0.002700%	0.000117	0.381327609	0.0781%	2034.4195	2035	0000000011110011	0.381550856	24.5786103	3%	
25.4	0.388151924	-0.000194	0.002700%	0.000147	0.388461019	0.0796%	2072.4769	2071	0000000010000011	0.38830065	24.6596078	3%	
25.5	0.395301412	-0.000193	0.002800%	0.000178	0.395622822	0.0813%	2110.6858	2115	0000000010000010	0.396550398	24.7586048	3%	
25.6	0.402478435	-0.000193	0.002800%	0.000208	0.402813208	0.0832%	2149.0472	2148	0000000010000100	0.402737709	24.8328525	3%	
25.7	0.410609553	-0.000192	0.002900%	0.000239	0.410957758	0.0848%	2192.4992	2191	0000000010000111	0.410799963	24.9295996	3%	
25.8	0.417840671	-0.000192	0.002900%	0.000269	0.418203152	0.0868%	2231.154	2230	0000000010010101	0.41811224	25.0173469	3%	
25.9	0.42509924	-0.000191	0.003000%	0.000300	0.425457866	0.0886%	2269.9546	2275	0000000010010001	0.426549482	25.1185938	3%	
26	0.432385229	-0.000191	0.003000%	0.000330	0.432777026	0.0906%	2309.907	2310	0000000010000010	0.433111782	25.1973414	3%	
26.1	0.43969861	-0.000190	0.003100%	0.000361	0.440105467	0.0925%	2348.005	2349	0000000010010010	0.440236655	25.2828388	3%	
26.2	0.447961449	-0.000189	0.003100%	0.000391	0.448383744	0.0943%	2392.1704	2395	0000000010010101	0.449048796	25.3858585	3%	
26.3	0.455328618	-0.000189	0.003200%	0.000422	0.455767927	0.0965%	2431.5657	2434	0000000010010000	0.456361073	25.4763239	3%	
26.4	0.462732084	-0.000188	0.003200%	0.000452	0.463178415	0.0984%	2471.1014	2474	0000000010010101	0.463860044	25.5663001	3%	
26.5	0.470144816	-0.000188	0.003300%	0.000483	0.470618097	0.1007%	2510.7928	2512	0000000010010100	0.470985266	25.6518275	3%	
26.6	0.477593781	-0.000187	0.003300%	0.000513	0.478094069	0.1026%	2550.6242	2551	0000000010011110	0.478297903	25.7395748	3%	
26.7	0.485069947	-0.000186	0.003400%	0.000544	0.48557807	0.1048%	2590.6057	2589	0000000010000011	0.485422686	25.8250722	3%	
26.8	0.49257328	-0.000186	0.003400%	0.000574	0.493102286	0.1070%	2630.7374	2629	0000000010010001	0.492924257	25.9150695	3%	
26.9	0.500103746	-0.000185	0.003500%	0.000605	0.500649604	0.1091%	2671.0138	2674	0000000010010100	0.501359699	26.0163164	3%	
27	0.507661311	-0.000185	0.003500%	0.000635									

28.1	0.591668042	-0.000178	0.004100%	0.000971	0.594479283	0.1371%	3160.9339	3193	0000110001001011	0.593044401	27.1165328	3%	0.983467
28.2	0.59865126	-0.000178	0.004100%	0.001001	0.599481410	0.1397%	3198.2986	3199	0000110001011111	0.599794195	27.1975303	4%	1.002470
28.3	0.606553484	-0.000177	0.004200%	0.001032	0.610315277	0.1421%	3240.6188	3240	0000110010100100	0.607481461	27.2897755	4%	1.010222
28.4	0.614488409	-0.000177	0.004200%	0.001062	0.615747678	0.1446%	3283.0948	3287	0000110010101011	0.616293692	27.3955743	4%	1.004476
28.5	0.622449859	-0.000176	0.004300%	0.001093	0.623395134	0.1470%	3325.7129	3327	0000110011011111	0.623793463	27.4855216	4%	1.014478
28.6	0.630437792	-0.000176	0.004300%	0.001123	0.631390797	0.1496%	3368.4771	3373	0000110101001010	0.6342182	27.5890184	4%	1.010982
28.7	0.637553119	-0.000175	0.004400%	0.001154	0.638522908	0.1521%	3406.5811	3409	0000110101010000	0.638909490	27.667766	4%	1.032234
28.8	0.645594828	-0.000174	0.004400%	0.001184	0.646592235	0.1545%	3449.6322	3449	0000110101011011	0.646667769	27.7601032	4%	1.039987
28.9	0.65366289	-0.000174	0.004500%	0.001215	0.654690268	0.1572%	3492.8355	3494	0000110101001010	0.655105007	27.8612601	4%	1.038740
29	0.661757259	-0.000173	0.004500%	0.001245	0.662819139	0.1596%	3536.1732	3538	0000110101101010	0.663534755	27.9602571	4%	1.039743
29.1	0.668982628	-0.000173	0.004600%	0.001276	0.670068561	0.1623%	3574.8802	3575	0000110110111011	0.670292044	28.0435405	4%	1.056495
29.2	0.677130246	-0.000172	0.004600%	0.001306	0.67824614	0.1648%	3618.5084	3622	0000110100010010	0.679104275	28.1492513	4%	1.050749
29.3	0.685304397	-0.000172	0.004700%	0.001337	0.686452106	0.1675%	3662.288	3667	0000110100100101	0.686416552	28.2639986	4%	1.063001
29.4	0.69261208	-0.000171	0.004700%	0.001367	0.693789411	0.1700%	3701.4332	3702	0000110101101011	0.694103817	28.3292458	4%	1.070754
29.5	0.700839208	-0.000170	0.004800%	0.001398	0.702044591	0.1726%	3745.4967	3749	0000110101001000	0.702728554	28.4327426	4%	1.067257
29.6	0.709092369	-0.000170	0.004800%	0.001428	0.710334684	0.1752%	3789.7036	3788	0000110101001011	0.710415819	28.5249988	4%	1.075010
29.7	0.716481964	-0.000169	0.004900%	0.001459	0.717556066	0.1778%	3829.2977	3829	0000110110111010	0.717915558	28.6149871	4%	1.085013
29.8	0.724788002	-0.000169	0.004900%	0.001489	0.726096073	0.1805%	3873.7924	3873	0000110100010000	0.726165339	28.7139841	4%	1.086016
29.9	0.732232296	-0.000168	0.005000%	0.001520	0.733573239	0.1831%	3913.6838	3914	0000110101001010	0.733852604	28.8062312	4%	1.093769
30	0.740591019	-0.000168	0.005000%	0.001550	0.741966978	0.1858%	3958.4652	3963	0000110101101110	0.743039824	28.9164779	4%	1.083522
30.1	0.748098623	-0.000167	0.005100%	0.001581	0.749496989	0.1885%	3998.6529	4000	0000110110100000	0.749977112	28.9997253	4%	1.100275
30.2	0.755616293	-0.000166	0.005100%	0.001611	0.757059823	0.1910%	4038.9969	4039	0000110111000011	0.757289389	29.0874727	4%	1.112527
30.3	0.764054516	-0.000166	0.005200%	0.001642	0.765535073	0.1938%	4084.2032	4085	0000110111101101	0.765914125	29.1909695	4%	1.109030
30.4	0.771634854	-0.000165	0.005200%	0.001672	0.773150464	0.1964%	4124.8312	4123	0001000000000010	0.773038908	29.2764669	4%	1.123533
30.5	0.779243069	-0.000165	0.005300%	0.001703	0.780795666	0.1992%	4165.6198	4166	0001000000000010	0.781101162	29.3732139	4%	1.126786
30.6	0.7875979631	-0.000164	0.005300%	0.001733	0.789344473	0.2018%	4211.2553	4211	0001000000001010	0.78935091	29.4722109	4%	1.127789
30.7	0.795421774	-0.000163	0.005400%	0.001764	0.797048665	0.2045%	4252.3133	4251	0001000000001011	0.797038176	29.5644581	4%	1.135542
30.8	0.803990217	-0.000163	0.005400%	0.001794	0.806556463	0.2072%	4298.2547	4302	0001000000001010	0.806600384	29.6792046	4%	1.120795
30.9	0.811706805	-0.000162	0.005500%	0.001825	0.813410453	0.2100%	4339.623	4338	0001000000001010	0.813350178	29.7602021	4%	1.139798
31	0.820326202	-0.000162	0.005500%	0.001855	0.821761036	0.2127%	4385.828	4388	0001000000001000	0.822724892	29.8726987	4%	1.127301
31.1	0.828995583	-0.000161	0.005600%	0.001886	0.828976626	0.2154%	4427.4876	4432	0001000000001000	0.83097464	29.9716957	4%	1.128304
31.2	0.835892131	-0.000161	0.005600%	0.001916	0.837716084	0.2182%	4469.2959	4468	0001000000001010	0.837724434	30.0526932	4%	1.147307
31.3	0.84458984	-0.000160	0.005700%	0.001947	0.844655512	0.2209%	4515.923	4517	0001000000001001	0.846911654	30.1623986	4%	1.137060
31.4	0.852439586	-0.000159	0.005700%	0.001977	0.854345572	0.2236%	4558.0154	4559	0001000000001011	0.85478613	30.257437	4%	1.142563
31.5	0.860316339	-0.000159	0.005800%	0.002008	0.86226645	0.2265%	4600.2648	4599	0001000000001101	0.862286184	30.3474342	4%	1.152566
31.6	0.869962248	-0.000158	0.005800%	0.002038	0.871595175	0.2291%	4651.9647	4651	0001000000000010	0.873160852	30.4779302	4%	1.122070
31.7	0.877890868	-0.000158	0.005900%	0.002069	0.879927323	0.2320%	4694.4969	4691	0001000000000001	0.880660624	30.5679275	4%	1.132073
31.8	0.885846367	-0.000157	0.005900%	0.002099	0.887925	0.2346%	4737.1652	4747	0001000000000010	0.888910372	30.6669245	4%	1.133076
31.9	0.893828637	-0.000157	0.006000%	0.002130	0.895951891	0.2375%	4779.9895	4783	0001000000001011	0.896785131	30.7614216	4%	1.138578
32	0.902704792	-0.000156	0.006000%	0.002160	0.904873193	0.2402%	4827.5855	4828	0001000000001010	0.905222374	30.8626685	4%	1.137332
32.1	0.910739479	-0.000155	0.006100%	0.002191	0.912562831	0.2430%	4870.6901	4913	0001000000000001	0.913659616	30.9639154	4%	1.136085
32.2	0.918800785	-0.000155	0.006100%	0.002221	0.921058928	0.2458%	4913.9401	4915	0001000000001001	0.921534376	31.0584125	4%	1.141587
32.3	0.926888652	-0.000154	0.006200%	0.002252	0.929133064	0.2486%	4957.3343	4958	0001000000001011	0.92959663	31.1551596	4%	1.144840
32.4	0.935003022	-0.000154	0.006200%	0.002282	0.937353931	0.2514%	5000.8733	5002	0001000000000010	0.937846378	31.2541565	4%	1.145843
32.5	0.944006025	-0.000153	0.006300%	0.002313	0.944605927	0.2542%	5044.1666	5050	0001000000001010	0.946846104	31.3621352	4%	1.137847
32.6	0.952172229	-0.000153	0.006300%	0.002343	0.954619768	0.2570%	5092.9882	5095	0001000000001101	0.955283346	31.4634002	4%	1.136600
32.7	0.959504569	-0.000152	0.006400%	0.002374	0.961989215	0.2599%	5132.353	5134	0001000000000010	0.962595623	31.5511475	4%	1.148853
32.8	0.967724366	-0.000151	0.006400%	0.002404	0.970265476	0.2626%	5176.461	5179	0001000000000101	0.971032866	31.6523944	3%	1.147606
32.9	0.975970371	-0.000151	0.006500%	0.002435	0.978682028	0.2655%	5220.7225	5223	0001000000000011	0.979282614	31.7513914	3%	1.148609
33	0.984242523	-0.000150	0.006500%	0.002465	0.986683047	0.2683%	5265.1159	5269	0001000000000101	0.987907351	31.8548882	3%	1.145112
33.1	0.992540761	-0.000150	0.006600%	0.002496	0.995232717	0.2712%	5309.6622	5308	0001000000000110	0.995219627	31.9426355	3%	1.157364
33.2	1.000865022	-0.000149	0.006600%	0.002526	1.000696987	0.2740%	5354.3398	5358	0001000000001011	1.004594341	32.0551321	3%	1.144688
33.3	1.009215245	-0.000149	0.006700%	0.002557	1.012098096	0.2769%	5399.1696	5401	0001000000000010	1.012656599	32.1518197	3%	1.148121
33.4	1.017591368	-0.000148	0.006700%	0.002587	1.020347088	0.2797%	5444.13	5445	0001000000000001	1.02096343	32.2508761	3%	1.149124
33.5	1.025993327	-0.000147	0.006800%	0.002618	1.028891814	0.2825%	5489.2368	5489	0001000000001011	1.029156092	32.3498731	3%	1.150127
33.6	1.034210558	-0.000147	0.006800%	0.002648	1.037372868	0.2854%	5534.484	5535	0001000000001011	1.037780828	32.4533699	3%	1.146630
33.7	1.042874498	-0.000146	0.006900%	0.002679	1.045882506	0.2882%	5579.8717	5584	0001000000000000	1.046968048	32.5636166	3%	1.136383
33.8	1.051353582	-0.000146	0.006900%	0.002709	1.054413832	0.2911%	5625.3992	5625	0001000000001011	1.054655313	32.6558638	3%	1.144136
33.9	1.059885825	-0.000145	0.007000%	0.002740	1.062973034	0.2939%	5671.0665	5673	0001000000000010	1.063655039	32.7639605	3%	1.136140
34	1.067541328	-0.000145	0.007000%	0.002770	1.071001305	0.2968%	5712.341	5717	0001000000000101	1.07190479	32.8628574	3%	1.137143
34.1	1.076097968	-0.000144	0.007100%	0.002801	1.079322908	0.2997%	5758.2919	5761	0001000000000001	1.080154535	32.9615844	3%	1.138146
34.2	1.084679989	-0.000143	0.007100%	0.002831	1.087960842	0.3025%	5804.3757	5803	0001000000000101	1.088029295	33.0635515	3%	1.143648
34.3	1.093287325	-0.000143	0.007200%	0.002862	1.096626704	0.3054%	5850.6089	5854	0001000000000110	1.097591503	33.171098	3%	1.128902
34.4	1.101919908	-0.000142	0.007200%	0.002892	1.105316315	0.3082%	5896.9688	5900	0001000000000001	1.10621624	33.2745949	3%	1.125405
34.5	1.110577672	-0.000142	0.007300%	0.002923	1.114033833	0.3112%	5943.477	5944	0001000000000010	1.114465988	33.3735919	3%	1.126408
34.6	1.118419569	-0.000141	0.007300%	0.002953	1.121931462	0.3140%	5985.6122	5985	0001000000000001	1			

	36.3	1.263232066	-0.000131	0.008200%	0.003472	1.267822745	0.3634%	6763.9562	6768	0001101001110000	1.268961273	35.2275353	3%	1.072465
	36.4	1.272364096	-0.000131	0.008200%	0.003502	1.277025034	0.3663%	6811.0513	6817	0001101010100001	1.278148493	35.3377819	3%	1.062218
	36.5	1.280698525	-0.000130	0.008300%	0.003533	1.285427736	0.3693%	6857.8806	6856	0001101011001000	1.28546077	35.4255292	3%	1.074471
	36.6	1.289058699	-0.000130	0.008300%	0.003563	1.293835653	0.3722%	6902.8488	6901	0001101011111011	1.293898012	35.5267761	3%	1.073224
	36.7	1.298263789	-0.000129	0.008400%	0.003594	1.303134044	0.3751%	6952.3454	6952	0001101100101000	1.30346022	35.6415226	3%	1.058477
	36.8	1.306674182	-0.000128	0.008400%	0.003624	1.311613258	0.3780%	6997.5828	6998	0001101101010110	1.312084957	35.7450195	3%	1.054981
	36.9	1.3151110094	-0.000128	0.008500%	0.003655	1.320120924	0.3810%	7042.9721	7042	0001101110000010	1.320334705	35.8440165	3%	1.055984
	37	1.324387549	-0.000127	0.008500%	0.003685	1.329471437	0.3839%	7092.8579	7092	0001101110110100	1.329709419	35.956513	3%	1.043487
	37.1	1.332873218	-0.000127	0.008600%	0.003716	1.338030123	0.3869%	7138.5194	7140	00011011111100100	1.338709145	36.0645097	3%	1.035490
	37.2	1.340570175	-0.000126	0.008600%	0.003746	1.345795307	0.3898%	7179.9474	7182	0001110000001110	1.346583904	36.1590069	3%	1.040993
	37.3	1.349107395	-0.000126	0.008700%	0.003777	1.35440677	0.3928%	7225.8903	7229	0001110000111101	1.355396135	36.2647536	3%	1.035246
	37.4	1.358481641	-0.000125	0.008700%	0.003807	1.363856623	0.3957%	7276.3062	7278	0001110001101110	1.364583355	36.3750003	3%	1.025000
	37.5	1.367067994	-0.000124	0.008800%	0.003838	1.372517493	0.3986%	7322.5128	7324	0001110010011100	1.373208092	36.4784971	3%	1.021503
	37.6	1.375679321	-0.000124	0.008800%	0.003868	1.381203592	0.4016%	7368.854	7368	0001110011001000	1.38145784	36.5774941	3%	1.022506
	37.7	1.384315544	-0.000123	0.008900%	0.003899	1.389915599	0.4045%	7415.3334	7416	0001110011111000	1.390457565	36.6854908	3%	1.014509
	37.8	1.393784257	-0.000123	0.008900%	0.003929	1.399463522	0.4075%	7466.2724	7470	0001110100101110	1.400582256	36.8069871	3%	0.993013
	37.9	1.402468973	-0.000122	0.009000%	0.003960	1.408225316	0.4104%	7513.0175	7513	0001110101011001	1.40864451	36.9037341	3%	0.996266
	38	1.411178344	-0.000122	0.009000%	0.003990	1.417011199	0.4134%	7559.8952	7562	0001110110001010	1.41783173	37.0139808	3%	0.986019
	38.1	1.419912286	-0.000121	0.009100%	0.004021	1.425824293	0.4164%	7606.9097	7608	0001110110111000	1.426456467	37.1174776	3%	0.982522
	38.2	1.429474163	-0.000120	0.009100%	0.004051	1.435467095	0.4192%	7656.3545	7662	0001110111101110	1.436581158	37.2389739	3%	0.961026
	38.3	1.438255946	-0.000120	0.009200%	0.004082	1.444329462	0.4223%	7705.6365	7707	0001110000101011	1.4450184	37.3402208	3%	0.959779
	38.4	1.447062055	-0.000119	0.009200%	0.004112	1.453214435	0.4252%	7753.0387	7754	0001110010010100	1.453830631	37.4459676	2%	0.954032
	38.5	1.455892409	-0.000119	0.009300%	0.004143	1.46212676	0.4282%	7800.5868	7805	0001110011111101	1.463392839	37.5607141	2%	0.939286
	38.6	1.464746923	-0.000118	0.009300%	0.004173	1.47106142	0.4311%	7848.2541	7853	0001111010101011	1.472392565	37.6687108	2%	0.931289
	38.7	1.472827387	-0.000117	0.009400%	0.004204	1.479220788	0.4341%	7891.7851	7891	0001111011010011	1.479517347	37.7542082	2%	0.945792
	38.8	1.481731037	-0.000117	0.009400%	0.004234	1.488206859	0.4370%	7939.7267	7938	0001111000000010	1.488329578	37.8599459	2%	0.940045
	38.9	1.490658597	-0.000116	0.009500%	0.004265	1.497217984	0.4400%	7987.8019	7991	0001111100110111	1.498266775	37.9792013	2%	0.920799
	39	1.499609982	-0.000116	0.009500%	0.004295	1.50625309	0.4430%	8036.0051	8036	0001111101100100	1.506704018	38.0804482	2%	0.919552
	39.1	1.508585107	-0.000115	0.009600%	0.004326	1.515313101	0.4460%	8084.3411	8089	0001111110011001	1.516641214	38.1996649	2%	0.903035
	39.2	1.516791096	-0.000115	0.009600%	0.004356	1.523606048	0.4489%	8128.5559	8131	0001111111000011	1.524515974	38.2941917	2%	0.905808
	39.3	1.525814514	-0.000114	0.009700%	0.004387	1.532710259	0.4519%	8177.1566	8179	0001111111110011	1.533515699	38.4021884	2%	0.897812
	39.4	1.534070765	-0.000113	0.009700%	0.004417	1.54104832	0.4548%	8221.641	8223	0010000000001111	1.541765448	38.5011854	2%	0.898815
	39.5	1.543142133	-0.000113	0.009800%	0.004448	1.550208201	0.4579%	8270.5098	8271	0010000001001111	1.550765173	38.6091821	2%	0.890818
	39.6	1.552233681	-0.000112	0.009800%	0.004478	1.559389496	0.4608%	8319.4929	8320	0010000001000000	1.559952393	38.7194287	2%	0.880571
	39.7	1.56135471	-0.000112	0.009900%	0.004509	1.568597253	0.4639%	8368.6172	8367	0010000001011111	1.568764624	38.8251755	2%	0.874825
	39.8	1.569709386	-0.000111	0.009900%	0.004539	1.577036291	0.4668%	8413.6402	8417	0010000001100001	1.578139338	38.9376721	2%	0.863238
	39.9	1.57887454	-0.000111	0.010000%	0.004570	1.586292633	0.4698%	8463.0237	8468	00100001000010100	1.587701546	39.0524185	2%	0.847581
	40	1.588062653	-0.000110	0.010000%	0.004600	1.595570009	0.4727%	8512.5194	8516	00100000101000100	1.596701271	39.1604153	2%	0.839585
	40.1	1.596490504	-0.000109	0.010100%	0.004631	1.604085757	0.4757%	8557.9517	8557	00100000101101101	1.604388536	39.2526624	2%	0.847338
	40.2	1.605725342	-0.000109	0.010100%	0.004661	1.613412186	0.4787%	8607.7091	8606	0010000110011110	1.613575756	39.3629091	2%	0.837091
	40.3	1.614982872	-0.000108	0.010200%	0.004692	1.622762602	0.4817%	8657.5945	8656	0010000111010000	1.62295047	39.4754056	2%	0.824584
	40.4	1.623483103	-0.000108	0.010200%	0.004722	1.631352073	0.4847%	8703.4202	8708	00100010000000100	1.632700172	39.5924021	2%	0.807598
	40.5	1.632786822	-0.000107	0.010300%	0.004753	1.640750031	0.4877%	8753.5592	8754	0010001000110010	1.641324909	39.6958989	2%	0.804101
	40.6	1.641335221	-0.000107	0.010300%	0.004783	1.649388971	0.4907%	8799.6487	8798	0010001001011110	1.649574657	39.7948959	2%	0.805104
	40.7	1.650684768	-0.000106	0.010400%	0.004814	1.658834117	0.4937%	8850.0395	8851	00100010100010011	1.659511854	39.9141423	2%	0.785858
	40.8	1.659280977	-0.000105	0.010400%	0.004844	1.667521168	0.4966%	8896.3858	8901	0010001011000101	1.668886568	40.0266388	2%	0.773361
	40.9	1.668675989	-0.000105	0.010500%	0.004875	1.677014151	0.4997%	8947.0317	8947	0010001011110011	1.677511305	40.1301357	2%	0.769864
	41	1.677319644	-0.000104	0.010500%	0.004905	1.685749963	0.5026%	8993.6381	8996	0010001100100100	1.686698524	40.2403823	2%	0.759618
	41.1	1.686759755	-0.000104	0.010600%	0.004936	1.695289416	0.5057%	9044.532	9043	0010001101010011	1.695510756	40.3461291	2%	0.753871
	41.2	1.694679247	-0.000103	0.010600%	0.004966	1.70329854	0.5086%	9087.2615	9087	0010001101111111	1.703760504	40.4445126	2%	0.754874
	41.3	1.703395014	-0.000103	0.010700%	0.004997	1.712111143	0.5117%	9134.2776	9139	0010001101100111	1.713510206	40.5621225	2%	0.737878
	41.4	1.712903706	-0.000102	0.010700%	0.005027	1.721718493	0.5146%	9185.5337	9190	0010001111100110	1.723072414	40.676869	2%	0.723131
	41.5	1.721666002	-0.000101	0.010800%	0.005058	1.730577972	0.5176%	9232.7999	9234	0010010000010010	1.731322163	40.775866	2%	0.724134
	41.6	1.730451967	-0.000101	0.010800%	0.005088	1.73946105	0.5206%	9280.1919	9279	0010010000111111	1.739759405	40.8771129	2%	0.722287
	41.7	1.740027319	-0.000100	0.010900%	0.005119	1.749138864	0.5236%	9331.824	9333	0010010001110101	1.749884096	40.9986092	2%	0.701391
	41.8	1.748859255	-0.000100	0.010900%	0.005149	1.758069255	0.5266%	9379.4685	9382	0010010010100110	1.759071316	41.1088558	2%	0.691144
	41.9	1.758478215	-0.000099	0.011000%	0.005180	1.767792077	0.5297%	9431.3407	9432	0010010011011000	1.76844603	41.2213524	2%	0.678648
	42	1.767355744	-0.000099	0.011000%	0.005210	1.77676941	0.5326%	9479.2356	9479	0010010100000111	1.777258261	41.3270991	2%	0.672901
	42.1	1.776256472	-0.000098	0.011100%	0.005241	1.785771396	0.5357%	9527.2621	9526	0010010100101110	1.786070492	41.4328459	2%	0.667154
	42.2	1.785940674	-0.000097	0.011100%	0.005271	1.795559764	0.5386%	9579.484	9584	0010010101110000	1.79694516	41.5633419	2%	0.636658
	42.3	1.794886427	-0.000097	0.011200%	0.005302	1.804609157	0.5417%	9627.7634	9632	0010010110100000	1.805944885	41.6713386	1%	0.628661
	42.4	1.803855094	-0.000096	0.011200%	0.005332	1.813679323	0.5446%	9676.1536	9675	0010010111001011	1.814007139	41.7680857	1%	0.631914
	42.5	1.812846581	-0.000096	0.011300%	0.005363	1.822775817	0.5477%	9724.6842	9728	00100110000000000	1.823944336	41.887332	1%	0.612668
	42.6	1.821860791	-0.000095	0.011300%	0.005393	1.831898276	0.5507%	9773.3246	9773	00100100001010101	1.832381578	41.9885789	1%	0.611421
	42.7	1.830897626	-0.00											

	44.5	1.994377615	-0.000084	0.012300%	0.005973	2.006504117	0.6080%	10704.8924	10708	0010100111010100	2.007688728	44.0922647	1%	0.407735
	44.6	2.003096018	-0.000084	0.012300%	0.006003	2.01533573	0.6110%	10752.0099	10756	0010101000000100	2.0166688454	44.2002614	1%	0.399739
	44.7	2.011837264	-0.000083	0.012400%	0.006034	2.024191888	0.6141%	10799.2583	10800	0010101000110000	2.0249388202	44.2992584	1%	0.400742
	44.8	2.020601253	-0.000082	0.012400%	0.006064	2.033069435	0.6171%	10846.6209	10847	0010101001011111	2.033750433	44.4050052	1%	0.394995
	44.9	2.030118609	-0.000082	0.012500%	0.006095	2.042708564	0.6202%	10896.0466	10902	0010101010010110	2.044062618	44.5287514	1%	0.371249
	45	2.038926672	-0.000081	0.012500%	0.006125	2.05163156	0.6231%	10945.6516	10945	0010101011000001	2.052124872	44.6254985	1%	0.374502
	45.1	2.049214211	-0.000081	0.012600%	0.006156	2.062046814	0.6262%	11001.218	11005	00101010111111101	2.063374529	44.7604943	1%	0.395906
	45.2	2.058064832	-0.000080	0.012600%	0.006186	2.071013765	0.6292%	11049.0576	11054	00101010110010110	2.072561749	44.8707471	1%	0.329259
	45.3	2.067663996	-0.000080	0.012700%	0.006217	2.080737581	0.6323%	11100.9351	11105	001010101101100001	2.082123957	44.9854875	1%	0.314513
	45.4	2.076557867	-0.000079	0.012700%	0.006247	2.089749168	0.6352%	11149.0127	11152	0010101110010000	2.090936188	45.0912343	1%	0.308766
	45.5	2.085473762	-0.000078	0.012800%	0.006278	2.098785573	0.6383%	11197.2228	11202	0010101111000010	2.100310902	45.2037308	1%	0.296269
	45.6	2.095134571	-0.000078	0.012800%	0.006308	2.108571072	0.6413%	11249.4294	11248	0010101111110000	2.108935638	45.3072277	1%	0.292772
	45.7	2.104093093	-0.000077	0.012900%	0.006339	2.117651519	0.6444%	11297.8745	11296	0010110000100000	2.117935364	45.4152244	1%	0.284776
	45.8	2.113073326	-0.000077	0.012900%	0.006369	2.126753248	0.6474%	11346.4331	11350	0010110001010110	2.128060055	45.5367207	1%	0.263279
	45.9	2.122794843	-0.000076	0.013000%	0.006400	2.136602729	0.6505%	11398.981	11403	0010110010001011	2.137997252	45.655967	1%	0.244033
	46	2.131817076	-0.000076	0.013000%	0.006430	2.145747862	0.6535%	11447.7712	11448	0010110010111000	2.146434494	45.7572139	1%	0.242786
	46.1	2.140860705	-0.000075	0.013100%	0.006461	2.1549416245	0.6565%	11496.6854	11495	0010110011100111	2.155246725	45.8629607	1%	0.237039
	46.2	2.149925626	-0.000074	0.013100%	0.006491	2.164104452	0.6595%	11545.7053	11548	0010110100001100	2.165183922	45.9822071	0%	0.217793
	46.3	2.159011732	-0.000074	0.013200%	0.006522	2.173317732	0.6626%	11594.8591	11599	0010110101001111	2.17474613	46.0969536	0%	0.203046
	46.4	2.167404779	-0.000073	0.013200%	0.006552	2.181830754	0.6656%	11640.2769	11639	0010110101101111	2.182245901	46.1869508	0%	0.213049
	46.5	2.177247078	-0.000073	0.013300%	0.006583	2.191806358	0.6687%	11693.4977	11697	0010110110110001	2.193120569	46.3174468	0%	0.182553
	46.6	2.186396106	-0.000072	0.013300%	0.006613	2.201081474	0.6717%	11742.9813	11743	0010110111011111	2.201745306	46.4209437	0%	0.179056
	46.7	2.195656895	-0.000071	0.013400%	0.006644	2.210380277	0.6747%	11792.5913	11797	0010110000010101	2.211869997	46.54244	0%	0.157560
	46.8	2.20475634	-0.000071	0.013400%	0.006674	2.219699225	0.6778%	11842.3088	11847	0010111001000111	2.221244711	46.6549365	0%	0.145063
	46.9	2.213967332	-0.000070	0.013500%	0.006705	2.22904066	0.6808%	11892.1462	11897	0010111001111001	2.230619425	46.7674331	0%	0.132567
	47	2.223198765	-0.000070	0.013500%	0.006735	2.238402012	0.6838%	11942.0899	11946	0010111010101010	2.239806644	46.8776797	0%	0.122320
	47.1	2.232450531	-0.000069	0.013600%	0.006766	2.247785657	0.6869%	11992.1526	11997	0010111011011011	2.249368853	46.9924262	0%	0.107574
	47.2	2.241722524	-0.000069	0.013600%	0.006796	2.257188986	0.6899%	12042.3203	12047	0010111100001111	2.258743567	47.1049228	0%	0.095077
	47.3	2.250310476	-0.000068	0.013700%	0.006827	2.266905416	0.6930%	12088.8233	12088	0010111100111000	2.266430832	47.19717	0%	0.102830
	47.4	2.259623707	-0.000067	0.013700%	0.006857	2.275350387	0.6960%	12139.2131	12139	0010111101101011	2.27599304	47.3119165	0%	0.088084
	47.5	2.268956841	-0.000067	0.013800%	0.006888	2.284819277	0.6991%	12189.7305	12190	0010111110011110	2.285555248	47.4226663	0%	0.073337
	47.6	2.278309769	-0.000066	0.013800%	0.006918	2.294305373	0.7021%	12240.3398	12244	0010111111010100	2.295679939	47.5481593	0%	0.051841
	47.7	2.288382104	-0.000066	0.013900%	0.006949	2.30451981	0.7052%	12294.8348	12295	0011000000000111	2.305242147	47.6629058	0%	0.037094
	47.8	2.297074574	-0.000065	0.013900%	0.006979	2.313341983	0.7082%	12341.9019	12341	0011000000110101	2.313866884	47.7664026	0%	0.033597
	47.9	2.306486233	-0.000065	0.014000%	0.007010	2.322892301	0.7113%	12392.8538	12392	0011000001101000	2.323429092	47.8811491	0%	0.018851
	48	2.315917251	-0.000064	0.014000%	0.007040	2.332459357	0.7143%	12443.8949	12446	0011000010011110	2.333553783	48.0026454	0%	0.002645
	48.1	2.32467224	-0.000063	0.014100%	0.007071	2.341348505	0.7174%	12491.3194	12494	0011000011001110	2.342553509	48.1106421	0%	0.010642
	48.2	2.333444859	-0.000063	0.014100%	0.007101	2.350258363	0.7204%	12538.8543	12538	0011000011111010	2.350803257	48.2096391	0%	0.009639
	48.3	2.34293925	-0.000062	0.014200%	0.007132	2.359898958	0.7235%	12590.2377	12595	00110000100110011	2.361490431	48.3378852	0%	0.037885
	48.4	2.351756885	-0.000062	0.014200%	0.007162	2.368842116	0.7265%	12636.0004	12641	00110000101100001	2.370115168	48.441382	0%	0.041382
	48.5	2.361286386	-0.000061	0.014300%	0.007193	2.378513618	0.7296%	12688.5988	12688	00110000110010000	2.378927399	48.5471288	0%	0.047129
	48.6	2.370144865	-0.000061	0.014300%	0.007223	2.38750842	0.7326%	12737.587	12736	00110000111000000	2.387927124	48.6551255	0%	0.055125
	48.7	2.379712769	-0.000060	0.014400%	0.007254	2.397127971	0.7357%	12789.398	12788	0011000011110100	2.397676826	48.7721219	0%	0.072122
	48.8	2.388611651	-0.000059	0.014400%	0.007284	2.406255393	0.7387%	12837.6039	12836	0011001000100100	2.406676552	48.8801186	0%	0.080119
	48.9	2.398217515	-0.000059	0.014500%	0.007315	2.416007184	0.7418%	12889.6306	12891	0011001001010111	2.416988737	49.0038648	0%	0.103865
	49	2.407156358	-0.000058	0.014500%	0.007345	2.425084176	0.7448%	12938.0572	12941	00110010100010101	2.426363451	49.1163614	0%	0.116361
	49.1	2.416799739	-0.000058	0.014600%	0.007376	2.434874949	0.7479%	12990.292	12990	0011001010111110	2.435550671	49.226608	0%	0.126608
	49.2	2.425778099	-0.000057	0.014600%	0.007406	2.443992881	0.7509%	13038.937	13044	0011001011110100	2.445675362	49.3481043	0%	0.148104
	49.3	2.434776606	-0.000057	0.014700%	0.007437	2.453135141	0.7540%	13087.7118	13086	00110010110001110	2.453550121	49.4426015	0%	0.142601
	49.4	2.444475984	-0.000056	0.014700%	0.007467	2.462980628	0.7570%	13140.2385	13139	00110010101010011	2.463487318	49.5618478	0%	0.161848
	49.5	2.45351334	-0.000055	0.014800%	0.007498	2.472162177	0.7601%	13189.2229	13192	0011001110001000	2.473424515	49.6810942	0%	0.181094
	49.6	2.462570509	-0.000055	0.014800%	0.007528	2.481362771	0.7631%	13238.309	13238	0011001110110110	2.482049252	49.784591	0%	0.184591
	49.7	2.470969877	-0.000054	0.014900%	0.007559	2.489902613	0.7662%	13283.8698	13287	0011001111100111	2.491236471	49.8948377	0%	0.194838
	49.8	2.480743835	-0.000054	0.014900%	0.007589	2.499826576	0.7692%	13336.8151	13337	0011010000011001	2.500611185	50.0073342	0%	0.207334
	49.9	2.490535048	-0.000053	0.015000%	0.007620	2.509770054	0.7723%	13389.8645	13394	0011010001010010	2.511298359	50.1355803	0%	0.235580
	50	2.499669234	-0.000053	0.015000%	0.007650	2.519050526	0.7754%	13439.3768	13444	0011010010000100	2.520673073	50.2480769	0%	0.248077
	50.1	2.508822671	-0.000052	0.015100%	0.007681	2.528352501	0.7784%	13489.0037	13494	0011010010101010	2.530047787	50.3605734	1%	0.260573
	50.2	2.517995247	-0.000051	0.015100%	0.007711	2.537677186	0.7814%	13538.723	13543	0011010011100111	2.539235006	50.4708201	1%	0.270820
	50.3	2.527186848	-0.000051	0.015200%	0.007742	2.547014384	0.7846%	13588.5666	13589	00110101000010101	2.547859743	50.5743169	1%	0.274317
	50.4	2.536397362	-0.000050	0.015200%	0.007772	2.556373043	0.7876%	13638.496	13643	00110101010010011	2.557984434	50.6958132	1%	0.295813
	50.5	2.545626676	-0.000050	0.015300%	0.007803	2.565754808	0.7907%	13688.5486	13689	0011010101111001	2.566609171	50.7993101	1%	0.299310
	50.6	2.554874677	-0.000049	0.015300%	0.007833	2.575152391	0.7937%	13738.6856	13737	00110101010101001	2.575608896	50.9073068	1%	0.307307
	50.7	2.56414125	-0.000048											

	52.7	2.748043725	-0.000037	0.016400%	0.008474	2.771624983	0.8581%	14796.8858	14789	0011100111000101	2.772852877	53.2742345	1%	0.574235	
	52.8	2.757039689	-0.000036	0.016400%	0.008504	2.780780957	0.8611%	14833.7338	14840	0011100111111000	2.782415085	53.388981	1%	0.588981	
	52.9	2.766053919	-0.000036	0.016500%	0.008535	2.789959573	0.8643%	14884.7034	14885	0011101000100101	2.790852327	53.4902279	1%	0.590228	
	53	2.775086295	-0.000035	0.016500%	0.008565	2.799153548	0.8673%	14933.7533	14935	0011101001010111	2.800227041	53.6027245	1%	0.602724	
	53.1	2.784136703	-0.000035	0.016600%	0.008596	2.808369952	0.8704%	14982.9237	14986	0011101010001010	2.809789249	53.717471	1%	0.617471	
	53.2	2.793843728	-0.000034	0.016600%	0.008626	2.818245412	0.8734%	15035.6102	15035	0011101010111011	2.818976469	53.8271716	1%	0.627718	
	53.3	2.802928744	-0.000034	0.016700%	0.008657	2.827497891	0.8766%	15084.9731	15086	0011101011110110	2.828538677	53.9424641	1%	0.642464	
	53.4	2.812031443	-0.000033	0.016700%	0.008687	2.836764916	0.8796%	15134.4136	15133	0011101100001101	2.837350908	54.0482109	1%	0.648211	
	53.5	2.8217871	-0.000032	0.016800%	0.008718	2.846694073	0.8827%	15187.3866	15191	0011101101010111	2.848225576	54.1787069	1%	0.678707	
	53.6	2.83092371	-0.000032	0.016800%	0.008748	2.855997469	0.8857%	15237.0211	15238	0011101110000110	2.857037807	54.2844537	1%	0.684454	
	53.7	2.840077654	-0.000031	0.016900%	0.008779	2.865320796	0.8888%	15286.7619	15285	0011101110110101	2.865850039	54.3902005	1%	0.690200	
	53.8	2.849248813	-0.000031	0.016900%	0.008809	2.874660184	0.8919%	15336.5885	15336	0011101111110100	2.875412247	54.504947	1%	0.704947	
	53.9	2.858437073	-0.000030	0.017000%	0.008840	2.884019291	0.8950%	15386.5202	15385	0011110000011001	2.884599466	54.6151936	1%	0.715194	
	54	2.867642315	-0.000030	0.017000%	0.008870	2.893394207	0.8980%	15436.5363	15439	0011110001001111	2.894724157	54.7366899	1%	0.736690	
	54.1	2.876864423	-0.000029	0.017100%	0.008901	2.902788628	0.9011%	15486.6564	15487	0011110001111111	2.903723883	54.8446866	1%	0.744687	
	54.2	2.88610328	-0.000028	0.017100%	0.008931	2.912197595	0.9041%	15536.8542	15538	0011110010110010	2.913286091	54.9594331	1%	0.759433	
	54.3	2.894105618	-0.000028	0.017200%	0.008962	2.920363395	0.9073%	15580.4195	15583	0011110011011111	2.921723333	55.06068	1%	0.760680	
	54.4	2.903379823	-0.000027	0.017200%	0.008992	2.929809145	0.9103%	15630.8135	15629	0011110100000101	2.93034807	55.1641768	1%	0.764177	
	54.5	2.912670425	-0.000027	0.017300%	0.009023	2.939275989	0.9134%	15681.32	15682	0011110101000010	2.940285267	55.2834232	1%	0.783423	
	54.6	2.921977309	-0.000026	0.017300%	0.009053	2.94875587	0.9165%	15731.8961	15731	0011110101110011	2.949472486	55.3936698	1%	0.793670	
	54.7	2.930678179	-0.000025	0.017400%	0.009084	2.957627742	0.9196%	15779.2284	15784	0011110110101000	2.959409683	55.5129162	1%	0.812916	
	54.8	2.940018371	-0.000025	0.017400%	0.009114	2.967143387	0.9226%	15829.9953	15834	00111101110101010	2.968784397	55.6254128	2%	0.825413	
	54.9	2.949374493	-0.000024	0.017500%	0.009145	2.976677768	0.9257%	15880.8616	15881	0011110000001001	2.977596628	55.7311595	2%	0.831160	
	55	2.958746427	-0.000024	0.017500%	0.009175	2.986226523	0.9288%	15931.8056	15932	0011110000111100	2.987158836	55.845906	2%	0.845906	
	55.1	2.968134057	-0.000023	0.017600%	0.009206	2.995793801	0.9319%	15982.848	15981	0011110011001101	2.996346056	55.9561527	2%	0.856153	
	55.2	2.976920574	-0.000023	0.017600%	0.009236	3.004752937	0.9349%	16030.6458	16033	0011111010100001	3.006095758	56.0731491	2%	0.873149	
	55.3	2.98634034	-0.000022	0.017700%	0.009267	3.01435382	0.9381%	16081.8675	16085	0011111011010101	3.015845461	56.1901455	2%	0.890146	
	55.4	2.995160951	-0.000021	0.017700%	0.009297	3.023347473	0.9411%	16129.8495	16128	0011111100000000	3.023907715	56.2868926	2%	0.886893	
	55.5	3.004612383	-0.000021	0.017800%	0.009328	3.032982503	0.9442%	16181.2533	16180	0011111100110100	3.033657417	56.403889	2%	0.903889	
	55.6	3.013466616	-0.000020	0.017800%	0.009358	3.042011214	0.9472%	16229.4223	16234	0011111101101010	3.043782108	56.5253853	2%	0.925385	
	55.7	3.022949244	-0.000020	0.017900%	0.009389	3.051678913	0.9504%	16281.0004	16283	0011111110010111	3.052969328	56.6356319	2%	0.935632	
	55.8	3.031226505	-0.000019	0.017900%	0.009419	3.060126288	0.9534%	16326.068	16331	0011111111000101	3.061969053	56.7436286	2%	0.943629	
	55.9	3.040131925	-0.000019	0.018000%	0.009450	3.069212435	0.9566%	16374.5434	16375	0011111111110111	3.070218801	56.8426256	2%	0.942626	
	56	3.049054171	-0.000018	0.018000%	0.009480	3.078312019	0.9596%	16423.0906	16426	01000000000101010	3.07978101	56.9573721	2%	0.957372	
	56.1	3.058599965	-0.000017	0.018100%	0.009511	3.088044808	0.9627%	16475.016	16480	0100000001100000	3.089905701	57.0788684	2%	0.978868	
	56.2	3.067554415	-0.000017	0.018100%	0.009541	3.097179046	0.9657%	16523.748	16523	01000000010001011	3.097967955	57.1756155	2%	0.975615	
	56.3	3.077129995	-0.000016	0.018200%	0.009572	3.106943099	0.9689%	16575.8402	16580	01000000011000100	3.108655128	57.3038615	2%	1.003862	
	56.4	3.086116176	-0.000016	0.018200%	0.009602	3.116110512	0.9719%	16624.7492	16624	01000000011100000	3.116904877	57.4028585	2%	1.002859	
	56.5	3.095118592	-0.000015	0.018300%	0.009633	3.125297019	0.9750%	16673.7601	16676	01000000100100100	3.126654579	57.5198549	2%	1.019855	
	56.6	3.104137124	-0.000015	0.018300%	0.009663	3.134498219	0.9781%	16722.8494	16725	0100000101010101	3.135841799	57.6301016	2%	1.030102	
	56.7	3.113171653	-0.000014	0.018400%	0.009694	3.143718296	0.9812%	16772.0394	16776	01000000110001000	3.145404007	57.7448481	2%	1.044848	
	56.8	3.122222061	-0.000013	0.018400%	0.009724	3.1529518	0.9842%	16821.301	16825	01000000110111001	3.154591227	57.8550947	2%	1.050595	
	56.9	3.13128823	-0.000013	0.018500%	0.009755	3.162205984	0.9874%	16870.673	16871	01000000111100111	3.163215963	57.9585916	2%	1.058592	
	57	3.140370042	-0.000012	0.018500%	0.009785	3.171472331	0.9904%	16920.1098	16921	01000010000011001	3.172590677	58.0710881	2%	1.071088	
	57.1	3.149467378	-0.000012	0.018600%	0.009816	3.178075914	0.9936%	16969.6558	16968	01000010010010000	3.181402908	58.1768349	2%	1.076835	
	57.2	3.15858012	-0.000011	0.018600%	0.009846	3.190057859	0.9966%	17019.2654	17021	0100001001111101	3.191340105	58.2960813	2%	1.096081	
	57.3	3.167114368	-0.000011	0.018700%	0.009877	3.198777117	0.9997%	17065.7835	17065	0100001010101001	3.199589853	58.3950782	2%	1.095078	
	57.4	3.176258652	-0.000010	0.018700%	0.009907	3.208108811	1.0028%	17115.569	17119	0100001011011111	3.209714544	58.5165745	2%	1.116575	
	57.5	3.184826374	-0.000009	0.018800%	0.009938	3.216861972	1.0059%	17162.2679	17167	0100001100001111	3.21871427	58.6245712	2%	1.124571	
	57.6	3.194592256	-0.000009	0.018800%	0.009968	3.226823656	1.0089%	17215.4145	17220	0100001010000100	3.228651466	58.7438176	2%	1.143818	
	57.7	3.203781339	-0.000008	0.018900%	0.009999	3.236205579	1.0121%	17265.4679	17266	0100001101110010	3.237276203	58.8473144	2%	1.147314	
	57.8	3.212396753	-0.000008	0.018900%	0.010029	3.24500634	1.0151%	17312.4208	17316	010000110100100	3.246650917	58.959811	2%	1.159811	
	57.9	3.221616194	-0.000007	0.019000%	0.010060	3.254420077	1.0182%	17362.644	17366	0100001111010110	3.256025631	59.0723076	2%	1.172308	
	58	3.230850096	-0.000007	0.019000%	0.010090	3.263846724	1.0213%	17412.9361	17417	01000100000001001	3.265587839	59.1870541	2%	1.187054	
	58.1	3.239513216	-0.000006	0.019100%	0.010121	3.272699622	1.0244%	17460.1671	17463	0100010000101111	3.274212576	59.2905509	2%	1.190551	
	58.2	3.248776766	-0.000005	0.019100%	0.010151	3.282156346	1.0275%	17510.6197	17510	0100010001100110	3.283024807	59.3962977	2%	1.196298	
	58.3	3.257471456	-0.000005	0.019200%	0.010182	3.291043268	1.0306%	17558.0323	17559	0100010010010111	3.292212027	59.5065443	2%	1.206544	
	58.4	3.266182291	-0.000004	0.019200%	0.010212	3.299942714	1.0336%	17605.5117	17609	0100010011001001	3.30158674	59.6190409	2%	1.219041	
	58.5	3.275489964	-0.000004	0.019300%	0.010243	3.309450066	1.0368%	17656.2343	17658	0100010011111010	3.31077396	59.7292875	2%	1.229288	
	58.6	3.284231654	-0.000003	0.019300%	0.010273	3.318381795	1.0398%	17703.8859	17707	0100010100101011	3.31996118	59.8395342	2%	1.239534	
	58.7	3.294146453	-0.000002	0.019400%	0.010304	3.328502748	1.0429%	17757.8822	17756	0100010101011100	3.329148399	59.9497808	2%	1.249781	
	58.8	3.302339864	-0.000002	0.019400%	0.010334	3.336882609	1.0460%	17802.5896	17802	010001010001010	3.337773136	60.0532776	2%	1.253278	
	58.9	3.311127485	-0.000001	0.019500%											

60.9	3.488851994	0.000010	0.020500%	0.010975	3.527614931	1.11114	18820.1648	18825	0100100101000101	3.529579782	32.3549574	2%	1.454957
61	3.497490869	0.000011	0.020500%	0.011005	3.536595495	1.11114	18868.7599	18873	0100100101110001	3.538579508	32.4629541	2%	1.462954
61.1	3.50648962	0.000011	0.020600%	0.011036	3.545669909	1.11739	18916.4887	18917	0100100101110011	3.546829256	32.5169151	2%	1.461951
61.2	3.515605328	0.000012	0.020600%	0.011066	3.554585248	1.12039	18966.2146	18971	0100100100000111	3.556953947	32.6834474	2%	1.483447
61.3	3.524182798	0.000012	0.020700%	0.011097	3.563775359	1.12359	19013.0842	19013	0100100100100001	3.564828707	32.7779445	2%	1.477944
61.4	3.5327774952	0.000013	0.020700%	0.011127	3.572573179	1.12659	19060.0114	19062	0100100100110110	3.574015926	32.8881911	2%	1.488191
61.5	3.5413930552	0.000014	0.020800%	0.011158	3.581492149	1.12969	19110.0007	19112	0100100101010000	3.58339064	32.0006877	2%	1.500688
61.6	3.550550714	0.000014	0.020800%	0.011188	3.590767572	1.13279	19157.0902	19162	0100100101010110	3.592765354	33.1131842	2%	1.513184
61.7	3.559731972	0.000015	0.020900%	0.011219	3.600164567	1.13589	19207.2241	19206	0100100100000011	3.601051102	32.2121812	2%	1.512181
61.8	3.567833961	0.000015	0.020900%	0.011249	3.609456186	1.13899	19251.5235	19252	0100100101010100	3.609639839	33.3156781	2%	1.515678
61.9	3.57649681	0.000016	0.021000%	0.011280	3.617347171	1.14209	19298.8661	19297	0100100101100001	3.618077082	33.416925	2%	1.516925
62	3.58517365	0.000017	0.021000%	0.011310	3.626226677	1.14519	19346.2677	19346	0100100110010010	3.627264301	33.5271716	2%	1.527172
62.1	3.594400933	0.000017	0.021100%	0.011341	3.635679499	1.14829	19396.6997	19398	0100100111000010	3.637014004	33.644168	2%	1.544168
62.2	3.603110333	0.000018	0.021100%	0.011371	3.644592333	1.15139	19444.2505	19446	0100100111110110	3.646013729	33.7521647	2%	1.552165
62.3	3.611286935	0.000018	0.021200%	0.011402	3.652977924	1.15459	19488.9963	19490	0100100100000100	3.654263477	33.8511671	2%	1.551162
62.4	3.620101859	0.000019	0.021200%	0.011432	3.661991595	1.15759	19536.6951	19538	0100100100010010	3.662362303	33.9591584	2%	1.559158
62.5	3.628763583	0.000019	0.021300%	0.011463	3.670891123	1.16079	19584.5043	19588	0100100100000010	3.672637197	34.071655	2%	1.571655
62.6	3.638059215	0.000020	0.021300%	0.011493	3.680398332	1.16379	19635.263	19640	0100100100111000	3.682387619	34.1886514	2%	1.588651
62.7	3.646829647	0.000021	0.021400%	0.011524	3.689832492	1.16689	19683.2099	19686	0100100101100110	3.691012356	34.2921483	2%	1.592148
62.8	3.655613128	0.000021	0.021400%	0.011554	3.698390547	1.16999	19731.2158	19732	0100101001000100	3.699637093	34.3956451	2%	1.595645
62.9	3.663875341	0.000022	0.021500%	0.011585	3.706854579	1.17319	19776.2586	19777	0100100100100001	3.708074335	34.496892	2%	1.596892
63	3.672688099	0.000022	0.021500%	0.011615	3.71588	1.17619	19824.5818	19827	0100101001000001	3.717449049	34.609386	2%	1.609389
63.1	3.681508574	0.000023	0.021600%	0.011646	3.724923136	1.17939	19872.8242	19873	0100101010100001	3.726073786	34.712854	2%	1.612854
63.2	3.689281952	0.000023	0.021600%	0.011676	3.732991654	1.18239	19915.3891	19914	010010101010010	3.733760151	34.8051326	2%	1.605133
63.3	3.698132034	0.000024	0.021700%	0.011707	3.741972889	1.18559	19963.7838	19966	0100101011111110	3.743510754	34.922129	2%	1.622129
63.4	3.707523476	0.000025	0.021700%	0.011737	3.75158186	1.18859	20015.0837	20017	0100101000100001	3.753072923	35.0368755	2%	1.636876
63.5	3.71869121	0.000025	0.021800%	0.011768	3.76015895	1.19179	20060.7666	20060	0100100100100101	3.76132271	35.1358725	2%	1.635873
63.6	3.724228949	0.000026	0.021800%	0.011798	3.768723667	1.19479	20106.5031	20105	0100100101000101	3.769572458	35.2348695	2%	1.634869
63.7	3.73128752	0.000026	0.021900%	0.011829	3.776748232	1.19799	20155.1836	20159	0100100101011111	3.779697149	35.356365	2%	1.653666
63.8	3.741515555	0.000027	0.021900%	0.011859	3.78643837	1.20109	20201.714	20202	01001001010101010	3.787759403	35.4531128	2%	1.653113
63.9	3.750440035	0.000027	0.022000%	0.011890	3.795602099	1.20419	20249.8911	20251	0100100110000101	3.798946623	35.563395	2%	1.663395
64	3.75858333	0.000028	0.022000%	0.011920	3.804229063	1.20729	20295.9278	20298	01001001010001010	3.805758654	35.6691062	2%	1.669106
64.1	3.767280243	0.000029	0.022100%	0.011951	3.812876234	1.21039	20342.0613	20342	0100100101010110	3.814008602	35.7681032	2%	1.668103
64.2	3.776241381	0.000029	0.022100%	0.011981	3.821601806	1.21349	20391.0672	20390	01001010101010010	3.823008327	35.8760999	2%	1.676100
64.3	3.784694082	0.000030	0.022200%	0.012012	3.830736189	1.21659	20437.3459	20438	0100101010101010	3.832008053	35.9820496	2%	1.684097
64.4	3.793678736	0.000030	0.022200%	0.012042	3.83979685	1.21969	20486.4847	20486	01000000000000011	3.841007776	36.0909033	2%	1.684093
64.5	3.802156773	0.000031	0.022300%	0.012073	3.848647798	1.22289	20532.906	20536	0100000000011000	3.850382492	36.2045899	2%	1.704590
64.6	3.810647816	0.000031	0.022300%	0.012103	3.857359739	1.22589	20579.3851	20583	0100000000010011	3.859194723	36.3103367	2%	1.710337
64.7	3.818636104	0.000032	0.022400%	0.012134	3.86655616	1.22909	20623.1672	20624	01000000000000010	3.866881988	36.4025389	2%	1.702584
64.8	3.827153826	0.000033	0.022400%	0.012164	3.875305169	1.23209	20669.7906	20671	0100000001011111	3.87569422	36.5083306	2%	1.708331
64.9	3.835682002	0.000033	0.022500%	0.012195	3.883602641	1.23529	20716.5115	20718	0100000001010110	3.884506451	36.6140774	2%	1.714077
65	3.844227115	0.000034	0.022500%	0.012225	3.891862804	1.23829	20763.2766	20764	01000000000000010	3.893131177	36.7175742	2%	1.717574
65.1	3.852782448	0.000034	0.022600%	0.012256	3.900611633	1.24149	20810.1381	20812	01000000010001011	3.902318407	36.8278209	2%	1.727821
65.2	3.861350085	0.000035	0.022600%	0.012286	3.909403631	1.24459	20857.0425	20862	01000000010111110	3.911505627	36.9380675	2%	1.738068
65.3	3.86992991	0.000035	0.022700%	0.012317	3.918212823	1.24769	20904.4021	20905	01000000010101001	3.919567681	37.0348146	2%	1.734815
65.4	3.878013293	0.000036	0.022700%	0.012347	3.92651526	1.25079	20948.3364	20947	01000001000100101	3.92744264	37.1293117	2%	1.729312
65.5	3.886618161	0.000037	0.022800%	0.012378	3.935350206	1.25389	20995.4717	20995	0100000000000000011	3.936442366	37.2373084	2%	1.737308
65.6	3.895234995	0.000037	0.022800%	0.012408	3.944194717	1.25699	21042.658	21046	01000001000101010	3.94600454	37.3520459	2%	1.752055
65.7	3.903863303	0.000038	0.022900%	0.012439	3.953054626	1.26019	21089.9965	21091	01000001000100001	3.954441816	37.4533018	2%	1.753302
65.8	3.912503348	0.000038	0.022900%	0.012469	3.961923853	1.26319	21137.2447	21141	01000001000100010	3.96381655	37.5657984	2%	1.765798
65.9	3.920651431	0.000039	0.023000%	0.012500	3.970298436	1.26639	21181.9239	21189	01000001000101110	3.971503796	37.6680455	2%	1.758046
66	3.92931536	0.000040	0.023000%	0.012530	3.979191994	1.26939	21229.3714	21229	01000001000101010	3.980316027	37.7673923	2%	1.763792
66.1	3.937489113	0.000040	0.023100%	0.012561	3.987598494	1.27259	21274.2004	21276	0100000100000100010	3.989182858	37.8695391	2%	1.769539
66.2	3.945175597	0.000041	0.023100%	0.012591	3.995491917	1.27569	21316.3724	21317	01000001000000010	3.996815523	37.9617863	2%	1.761786
66.3	3.953875995	0.000041	0.023200%	0.012622	4.003436257	1.27889	21364.0524	21367	01000001000101010	4.006190237	38.0742828	2%	1.774283
66.4	3.962587315	0.000042	0.023200%	0.012652	4.011379528	1.28189	21411.7674	21412	01000001000101000	4.01642768	38.1755298	2%	1.775530
66.5	3.970812019	0.000042	0.023300%	0.012683	4.019383232	1.28509	21456.883	21463	01000001000101010	4.023814699	38.2857764	2%	1.785776
66.6	3.979049419	0.000043	0.023300%	0.012713	4.027306987	1.28809	21502.2408	21501	0000000000000000011	4.031689459	38.3802735	2%	1.780274
66.7	3.987794823	0.000044	0.023400%	0.012744	4.039284726	1.29129	21549.9724	21552	0100000000000000000	4.040876678	38.4905201	2%	1.790520
66.8	3.996057427	0.000044	0.023400%	0.012774	4.047775817	1.29439	21595.2732	21596	01000000000001010	4.049126427	38.5985171	2%	1.798517
66.9	4.004330067	0.000045	0.023500%	0.012805	4.056283026	1.29749	21640.66	21643	0100000000000000001	4.057938658	38.6952639	2%	1.795264
67	4.013108526	0.000045	0.023500%	0.012835	4.065298893	1.30059	21686.7605	21691	010000000000010101	4.066938383	38.8032606	2%	1.803261
67.1	4.02140568	0.000046	0.023600%	0.012866	4.073830841	1.30379	21734.2792	21735	0100000000000101011	4.075188131	38.9022576	2%	1.802258
67.2	4.029714839	0.000046	0.023600%	0.012896	4.082372365	1.30679	21779.8491	21782	0100000000000000000	4.084000362	39.0008043	2%	1.808004
67.3	4.038035887	0.000047	0.023700%	0.012927	4.090920554	1.30999	21825.5207						

	69.1	4.185943996	0.000057	0.024600%	0.013476	4.243124561	1.3660%	22637.4775	22636	0101100001101100	4.244120476	70.9294457	3%	1.829446
	69.2	4.194006373	0.000058	0.024600%	0.013506	4.25142534	1.3691%	22681.763	22685	0101100010001101	4.253307695	71.0396923	3%	1.839692
	69.3	4.201610495	0.000058	0.024700%	0.013537	4.259267413	1.3723%	22723.6012	22724	01011000011000100	4.260619972	71.1274397	3%	1.827440
	69.4	4.209696863	0.000059	0.024700%	0.013567	4.267593517	1.3753%	22768.0217	22768	0101100011110000	4.26886972	71.2264366	3%	1.826437
	69.5	4.21779457	0.000060	0.024800%	0.013598	4.275935959	1.3785%	22812.5295	22817	01011000100100001	4.27805694	71.3366833	3%	1.836683
	69.6	4.225903504	0.000060	0.024800%	0.013628	4.284268976	1.3816%	22857.0829	22856	01011000101001000	4.285369217	71.4244306	3%	1.824431
	69.7	4.234023552	0.000061	0.024900%	0.013659	4.292653108	1.3847%	22901.7171	22903	0101100010110111	4.294818448	71.5301774	3%	1.830177
	69.8	4.2421546	0.000061	0.024900%	0.013689	4.301027573	1.3878%	22946.3956	22949	0101100010100101	4.302806185	71.6336742	3%	1.833674
	69.9	4.249832509	0.000062	0.025000%	0.013720	4.308946458	1.3910%	22988.6436	22987	01011000111000101	4.309930967	71.7191716	3%	1.819172
	70	4.258449253	0.000063	0.025000%	0.013750	4.317813389	1.3940%	23033.9496	23035	01011000111111011	4.318930693	71.8271683	3%	1.827168
	70.1	4.266612633	0.000063	0.025100%	0.013781	4.326226562	1.3972%	23080.8347	23085	0101101000010101	4.328350407	71.9396649	3%	1.839665
	70.2	4.274325413	0.000064	0.025100%	0.013811	4.334177944	1.4003%	23123.2561	23125	0101101001010101	4.335805178	72.0296621	3%	1.829662
	70.3	4.282510746	0.000064	0.025200%	0.013842	4.342614412	1.4035%	23168.2654	23169	01011010100000001	4.344054926	72.1286591	3%	1.828659
	70.4	4.290247169	0.000065	0.025200%	0.013872	4.350590759	1.4065%	23210.82	23215	0101101010101011	4.352679663	72.232156	3%	1.832156
	70.5	4.298454004	0.000065	0.025300%	0.013903	4.359050071	1.4097%	23255.9512	23256	01011010110110100	4.360366928	72.3244031	3%	1.824403
	70.6	4.306670948	0.000066	0.025300%	0.013933	4.367514722	1.4128%	23301.111	23301	01011011000000101	4.368804171	72.42565	3%	1.825650
	70.7	4.314441509	0.000067	0.025400%	0.013964	4.375531622	1.4159%	23343.8819	23347	0101101100110011	4.377428907	72.5291469	3%	1.829147
	70.8	4.321768431	0.000067	0.025400%	0.013994	4.38309566	1.4190%	23384.2368	23388	01011011010101100	4.385116173	72.6213941	3%	1.821394
	70.9	4.330017884	0.000068	0.025500%	0.014025	4.39159925	1.4222%	23429.6042	23431	01011011100000111	4.393178427	72.7181411	3%	1.818141
	71	4.338277002	0.000068	0.025500%	0.014055	4.400109729	1.4253%	23475.0085	23478	0101101101101010	4.401990565	72.8238879	3%	1.823888
	71.1	4.346093091	0.000069	0.025600%	0.014086	4.408174824	1.4284%	23510.0365	23520	0101101111100000	4.409865417	72.918385	3%	1.818385
	71.2	4.353920518	0.000069	0.025600%	0.014116	4.416248462	1.4315%	23561.1102	23562	01011100000001010	4.417740177	73.0128821	3%	1.812882
	71.3	4.36175917	0.000070	0.025700%	0.014147	4.424337451	1.4347%	23604.2657	23609	01011100001110001	4.426552408	73.1186289	3%	1.818629
	71.4	4.369608935	0.000071	0.025700%	0.014177	4.432433726	1.4378%	23647.4601	23648	0101110001100000	4.433864865	73.2063762	3%	1.806376
	71.5	4.377918511	0.000071	0.025800%	0.014208	4.44100245	1.4410%	23693.1751	23698	0101110010010010	4.443239399	73.3188728	3%	1.818873
	71.6	4.385789227	0.000072	0.025800%	0.014238	4.449120986	1.4440%	23736.4882	23738	01011100101101010	4.45073917	73.40887	3%	1.808870
	71.7	4.393223792	0.000072	0.025900%	0.014269	4.456803087	1.4472%	23779.473	23779	01011100111000011	4.458426435	73.5011172	3%	1.801172
	71.8	4.401116893	0.000073	0.025900%	0.014299	4.464945317	1.4503%	23820.9126	23822	0101110100000110	4.466486869	73.5978643	3%	1.797864
	71.9	4.409020552	0.000073	0.026000%	0.014330	4.473104305	1.4535%	23864.4415	23869	01011101001110101	4.47530092	73.703611	3%	1.803611
	72	4.417378778	0.000074	0.026000%	0.014360	4.481719436	1.4565%	23910.4041	23910	0101110101100100	4.482988186	73.7958582	2%	1.795858
	72.1	4.424859096	0.000075	0.026100%	0.014391	4.489448921	1.4597%	23951.6416	23950	0101110110000110	4.490487957	73.885855	2%	1.785855
	72.2	4.432793763	0.000075	0.026100%	0.014421	4.49763634	1.4628%	23995.3223	23996	01011101101101100	4.499112694	73.9893523	2%	1.789352
	72.3	4.440297222	0.000076	0.026200%	0.014452	4.505390284	1.4660%	24036.6903	24038	0101110111100100	4.506987453	74.0838494	2%	1.783849
	72.4	4.448252942	0.000076	0.026200%	0.014482	4.513600058	1.4691%	24080.4903	24082	0101110000010010	4.515237202	74.1828464	2%	1.782846
	72.5	4.455779093	0.000077	0.026300%	0.014513	4.52137801	1.4722%	24121.9864	24121	01011100001110001	4.522549479	74.2705937	2%	1.770594
	72.6	4.463755425	0.000077	0.026300%	0.014543	4.529609696	1.4753%	24165.9032	24169	01011110011010001	4.531549204	74.3785904	2%	1.778590
	72.7	4.471303823	0.000078	0.026400%	0.014574	4.537411209	1.4785%	24207.5251	24212	01011110100010100	4.539611458	74.4753375	2%	1.775337
	72.8	4.478863641	0.000079	0.026400%	0.014604	4.545220189	1.4815%	24249.1867	24254	0101111010111100	4.547486217	74.5698346	2%	1.769835
	72.9	4.48667053	0.000079	0.026500%	0.014635	4.553488992	1.4847%	24293.3016	24292	0101111011100100	4.554611	74.655332	2%	1.755332
	73	4.494886771	0.000080	0.026500%	0.014665	4.561762162	1.4878%	24337.4397	24340	01011111000010100	4.563610725	74.7633287	2%	1.763329
	73.1	4.502478335	0.000080	0.026600%	0.014696	4.569610478	1.4910%	24379.3113	24383	0101111100111111	4.571672979	74.8600758	2%	1.760076
	73.2	4.510080875	0.000081	0.026600%	0.014726	4.577464767	1.4941%	24421.2146	24425	01011111011010001	4.579547739	74.9545729	2%	1.754573
	73.3	4.518126366	0.000081	0.026700%	0.014757	4.585774789	1.4973%	24465.5494	24467	01011111100100011	4.587422499	75.04907	2%	1.749070
	73.4	4.52574961	0.000082	0.026700%	0.014787	4.593651071	1.5003%	24507.5701	24509	01011111101101101	4.595297258	75.1435671	2%	1.743567
	73.5	4.53383501	0.000083	0.026800%	0.014818	4.60154331	1.5035%	24549.676	24553	01011111110100001	4.603547007	75.2425641	2%	1.742564
	73.6	4.541027928	0.000083	0.026800%	0.014848	4.609443077	1.5066%	24591.822	24594	0110000000010010	4.611234272	75.3348113	2%	1.734811
	73.7	4.548254354	0.000084	0.026900%	0.014879	4.616922684	1.5098%	24631.7264	24635	0110000000110011	4.618921537	75.4270584	2%	1.727058
	73.8	4.556347954	0.000084	0.026900%	0.014909	4.62527938	1.5129%	24676.3102	24679	0110000001100111	4.627171286	75.5260554	2%	1.726055
	73.9	4.564023335	0.000085	0.027000%	0.014940	4.633215711	1.5160%	24718.6513	24719	0110000010001111	4.634671057	75.6160527	2%	1.716053
	74	4.570857421	0.000086	0.027000%	0.014970	4.64029375	1.5191%	24756.4133	24760	0110000010110000	4.642358322	75.7082999	2%	1.708300
	74.1	4.578554707	0.000086	0.027100%	0.015001	4.6482543	1.5223%	24798.8836	24801	0110000011000001	4.650045588	75.8005471	2%	1.700547
	74.2	4.586685761	0.000087	0.027100%	0.015031	4.656649996	1.5254%	24843.6755	24847	0110000100000111	4.658670324	75.9040439	2%	1.704044
	74.3	4.593978818	0.000087	0.027200%	0.015062	4.664201142	1.5286%	24883.9615	24886	0110000100101010	4.665982601	75.9917912	2%	1.691791
	74.4	4.601705426	0.000088	0.027200%	0.015092	4.6721872	1.5316%	24926.5679	24925	0110000101010101	4.673294878	76.0795385	2%	1.679539
	74.5	4.609020417	0.000088	0.027300%	0.015123	4.679761574	1.5348%	24966.9779	24969	01100001100001001	4.681544626	76.1785355	2%	1.678536
	74.6	4.616766937	0.000089	0.027300%	0.015153	4.687768819	1.5379%	25009.6974	25008	0110000110110000	4.688856903	76.2662828	2%	1.666283
	74.7	4.624522802	0.000090	0.027400%	0.015184	4.695790809	1.5411%	25052.4955	25051	0110000111010101	4.696919157	76.3630299	2%	1.663030
	74.8	4.631869424	0.000090	0.027400%	0.015214	4.703393978	1.5442%	25093.0591	25095	01100010000000111	4.705168905	76.4620269	2%	1.662027
	74.9	4.639226971	0.000091	0.027500%	0.015245	4.711012472	1.5474%	25133.7045	25138	0110001000110010	4.713231159	76.5587739	2%	1.658774
	75	4.647012023	0.000091	0.027500%	0.015275	4.719061806	1.5505%	25176.6485	25178	01100010010101010	4.72073093	76.6487712	2%	1.648771
	75.1	4.654390197	0.000092	0.027600%	0.015306	4.726702214	1.5536%	25217.4108	25219	01100010100000011	4.728481896	76.7410183	2%	1.641018
	75.2	4.66177897	0.000092	0.027600%	0.015336	4.734350073	1.5567%	25258.2128	25262	0110001010101100	4.73648045	76.8377654	2%	1.637765
	75.3	4.669178231	0.000093	0.027700%	0.015367	4.742012851	1.5599%	25299.0945	25298	0110001011010010	4.743230244	76.9187629	2%	1.618763
	75.4	4.677000998	0.000094	0.027700%	0.015397	4.750101417	1.5630%	25342						

77.3	4.816426028	0.001004	0.028700%	0.015977	4.894578803	1.6227%	26113.0539	26112	0.0110010000000000	4.898580596	78.750207	2%	1.450207
77.4	4.82323576	0.001005	0.028700%	0.016007	4.905169519	1.6257%	26115.7661	26112	0.0110010001000000	4.903350357	78.8420043	2%	1.440204
77.5	4.830846648	0.001006	0.028800%	0.016038	4.905366787	1.6289%	26129.8512	26195	0.0110010010001001	4.911421261	78.9369513	2%	1.436951
77.6	4.838071822	0.001006	0.028800%	0.016068	4.917029581	1.6320%	26232.8256	26234	0.0110010011110100	4.917248888	79.0246987	2%	1.424698
77.7	4.844913332	0.001007	0.028900%	0.016099	4.924137651	1.6352%	26270.7448	26269	0.0110010011001100	4.925287188	79.1034663	2%	1.403466
77.8	4.852158466	0.001007	0.028900%	0.016129	4.931611134	1.6383%	26310.8333	26312	0.0110010011001000	4.933349442	79.2001933	2%	1.400193
77.9	4.85941307	0.001008	0.029000%	0.016160	4.939179277	1.6415%	26350.9963	26350	0.0110010011011010	4.940474224	79.2856907	2%	1.385691
78	4.866677032	0.001009	0.029000%	0.016190	4.946712366	1.6446%	26391.1861	26393	0.0110010011000010	4.948536478	79.3824377	2%	1.382438
78.1	4.873560604	0.001009	0.029100%	0.016221	4.953865313	1.6478%	26429.3477	26428	0.0110010011001100	4.955098778	79.4611853	2%	1.361185
78.2	4.881232623	0.001010	0.029100%	0.016251	4.961814122	1.6508%	26471.7554	26473	0.0110010011010001	4.96353602	79.5624322	2%	1.362432
78.3	4.888136096	0.001010	0.029200%	0.016282	4.968898218	1.6540%	26505.1029	26515	0.0110010011000101	4.97141078	79.6569294	2%	1.356929
78.4	4.89505021	0.001011	0.029200%	0.016312	4.976167508	1.6571%	26548.3321	26552	0.0110010011011000	4.978348068	79.7401768	2%	1.340177
78.5	4.902381114	0.001011	0.029300%	0.016343	4.983756701	1.6603%	26588.8212	26592	0.011001001101100000	4.985847839	79.8301741	2%	1.330174
78.6	4.909860757	0.001012	0.029300%	0.016373	4.991348167	1.6634%	26629.3277	26632	0.0101000000000000	4.99334761	79.9201713	2%	1.320174
78.7	4.916624467	0.001013	0.029400%	0.016404	4.998564975	1.6666%	26677.8247	26671	0.0101000000101111	5.000659887	80.0079186	2%	1.307919
78.8	4.923962124	0.001013	0.029400%	0.016434	5.006177695	1.6697%	26708.4393	26707	0.0101000001000000	5.007409681	80.0889162	2%	1.288916
78.9	4.930925325	0.001014	0.029500%	0.016465	5.013414236	1.6729%	26747.4047	26749	0.0101000001100001	5.015284441	80.1834433	2%	1.283413
79	4.93789835	0.001014	0.029500%	0.016495	5.02065321	1.6760%	26785.6895	26789	0.0101000001111001	5.022784212	80.2734105	2%	1.273411
79.1	4.944500436	0.001015	0.029600%	0.016526	5.027527395	1.6792%	26822.342	26822	0.0101000010000010	5.028971523	80.3476583	2%	1.247658
79.2	4.95148184	0.001015	0.029600%	0.016556	5.034792245	1.6823%	26861.1007	26865	0.0101000011000010	5.037033777	80.4444053	2%	1.234405
79.3	4.958497626	0.001016	0.029700%	0.016587	5.042071504	1.6855%	26899.9363	26900	0.0101000010000100	5.043596077	80.5231529	2%	1.223153
79.4	4.965889368	0.001017	0.029700%	0.016617	5.049740948	1.6886%	26940.8535	26940	0.0101000010110000	5.051095848	80.6131502	2%	1.213150
79.5	4.972531575	0.001017	0.029800%	0.016648	5.056656426	1.6918%	26977.7482	26977	0.0101000010100001	5.058033136	80.6963976	2%	1.196398
79.6	4.979556076	0.001018	0.029800%	0.016678	5.063968687	1.6948%	27016.718	27016	0.0101000010000100	5.065345413	80.784145	2%	1.184145
79.7	4.986606256	0.001018	0.029900%	0.016709	5.071281327	1.6981%	27055.7735	27059	0.010100001010010001	5.073407667	80.880892	1%	1.168892
79.8	4.993656614	0.001019	0.029900%	0.016739	5.078605419	1.7011%	27094.8482	27097	0.010100001010010001	5.080532425	80.9663894	1%	1.160639
79.9	5.000341479	0.001019	0.030000%	0.016770	5.085564422	1.7043%	27131.9572	27132	0.0101000000000000	5.080322221	81.0563867	1%	1.156387
80	5.002267689	0.001020	0.030000%	0.016800	5.086776427	1.7074%	27173.2497	27177	0.0101000000000000	5.088969692	81.0676363	1%	1.067636